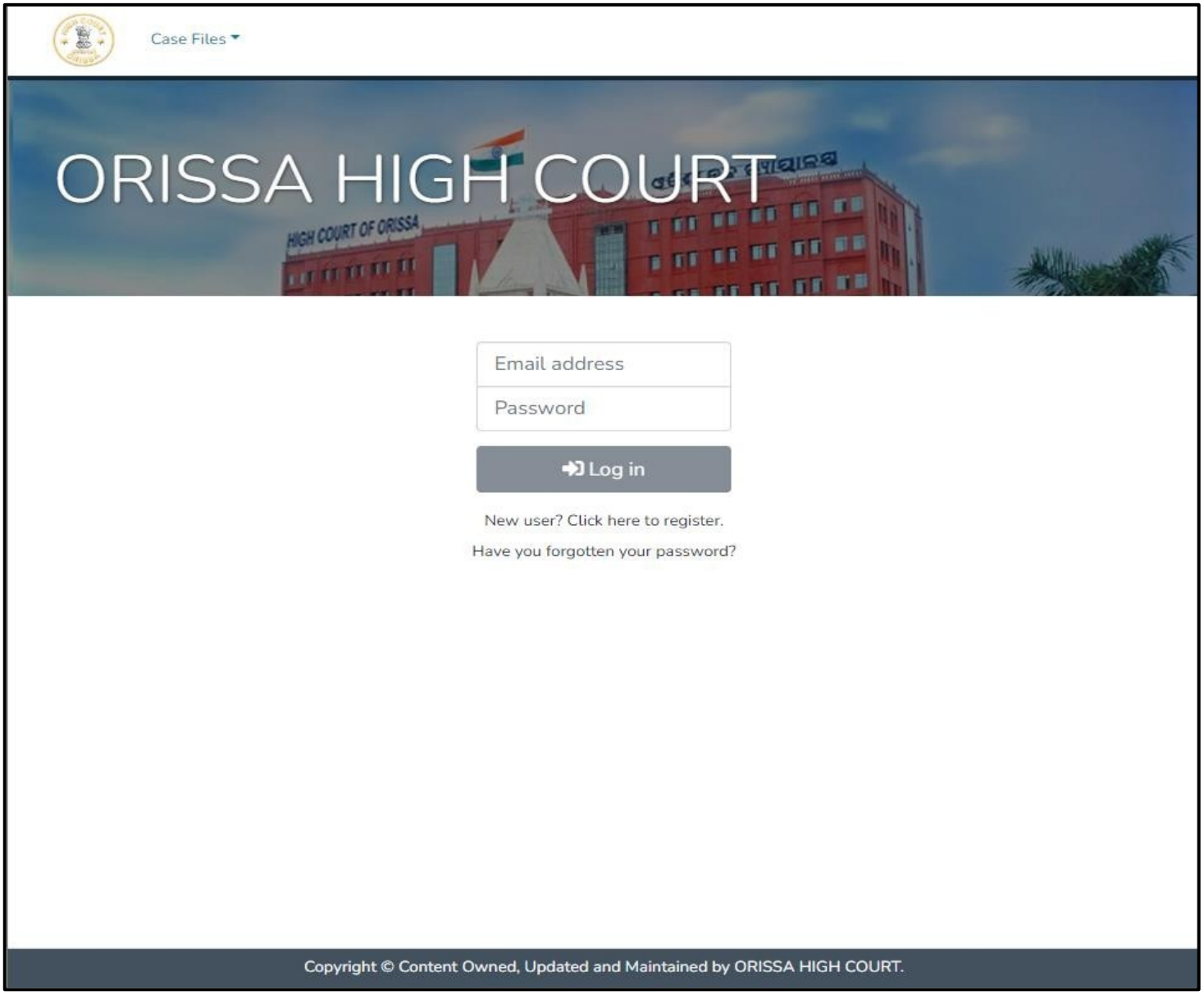
Step:1 Hierarchical File Storage Structure & UI Organization
Implement a hierarchical file storage structure within the DMS:

- **Judicial Record** (Top-Level Community)
  - **Civil & Criminal** (Sub-Communities)
    - **Items categorized as per court specifications.**

Home page

The DMS implementation home page contains three main sections: Search boxes and tabs for searching various case records by different parameters, admin and users sign on tabs for users and admin operations and self-registration or forgot password options.

Login page

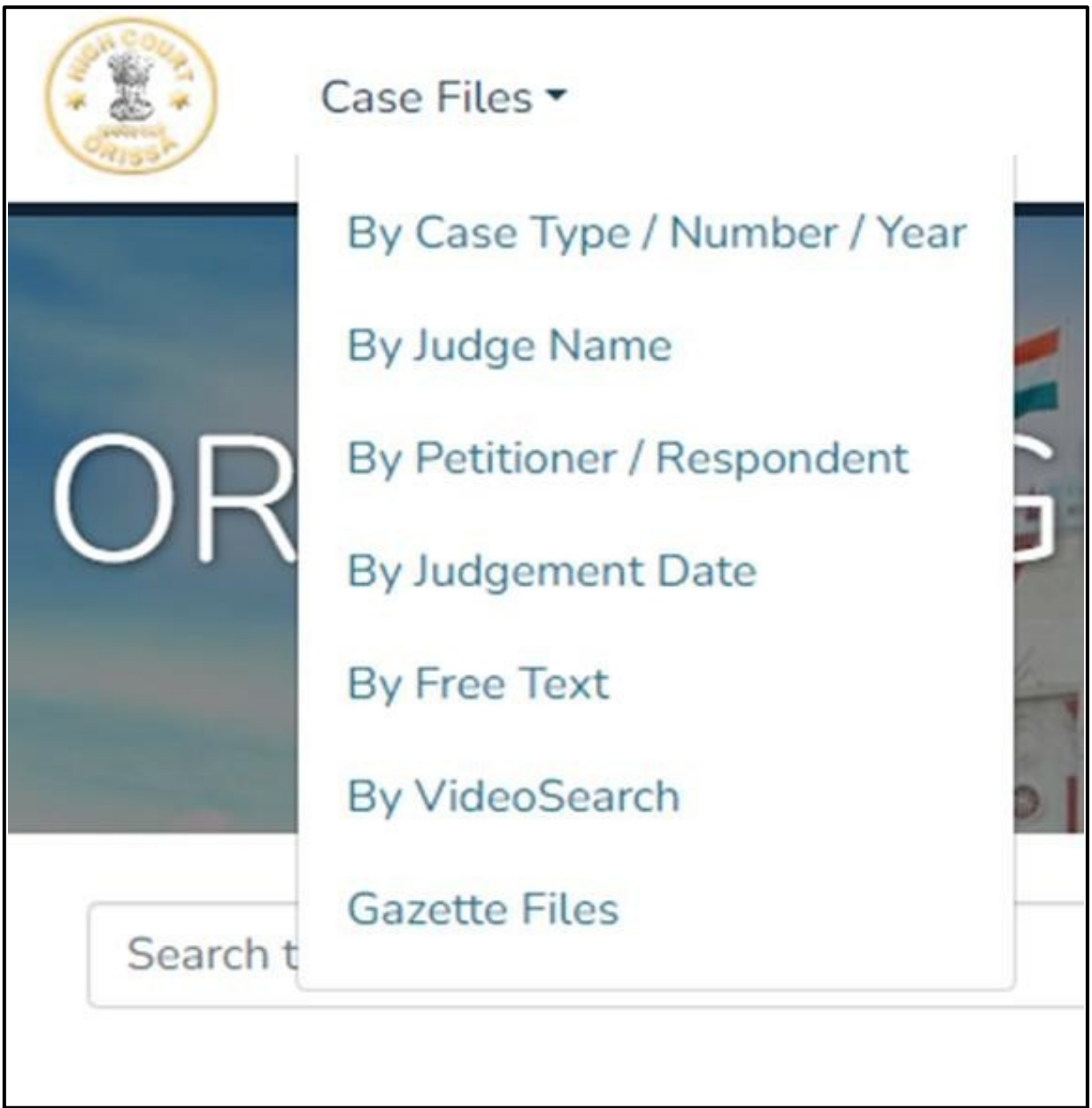


Using this page, the authorized user can login into the DMS and perform numerous functions and operations.

Search tab

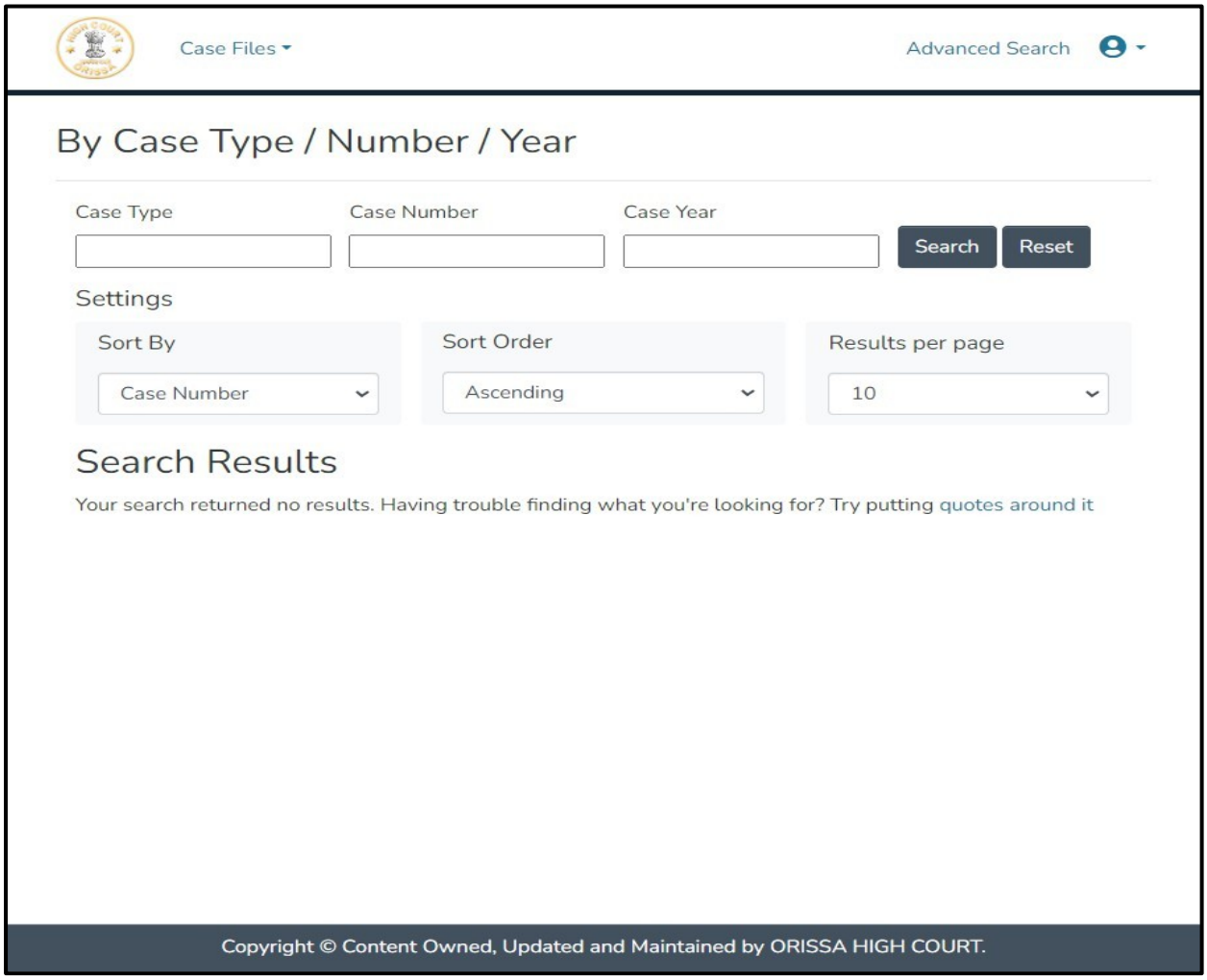
In search tab for All Case Files following parameters are defined:

1. By Case Type/No./Year
2. By Judge Name
3. By Petitioner/Respondent
4. By Judgement Date
5. By Free Text Search
6. By Video Search
7. Gazette Files



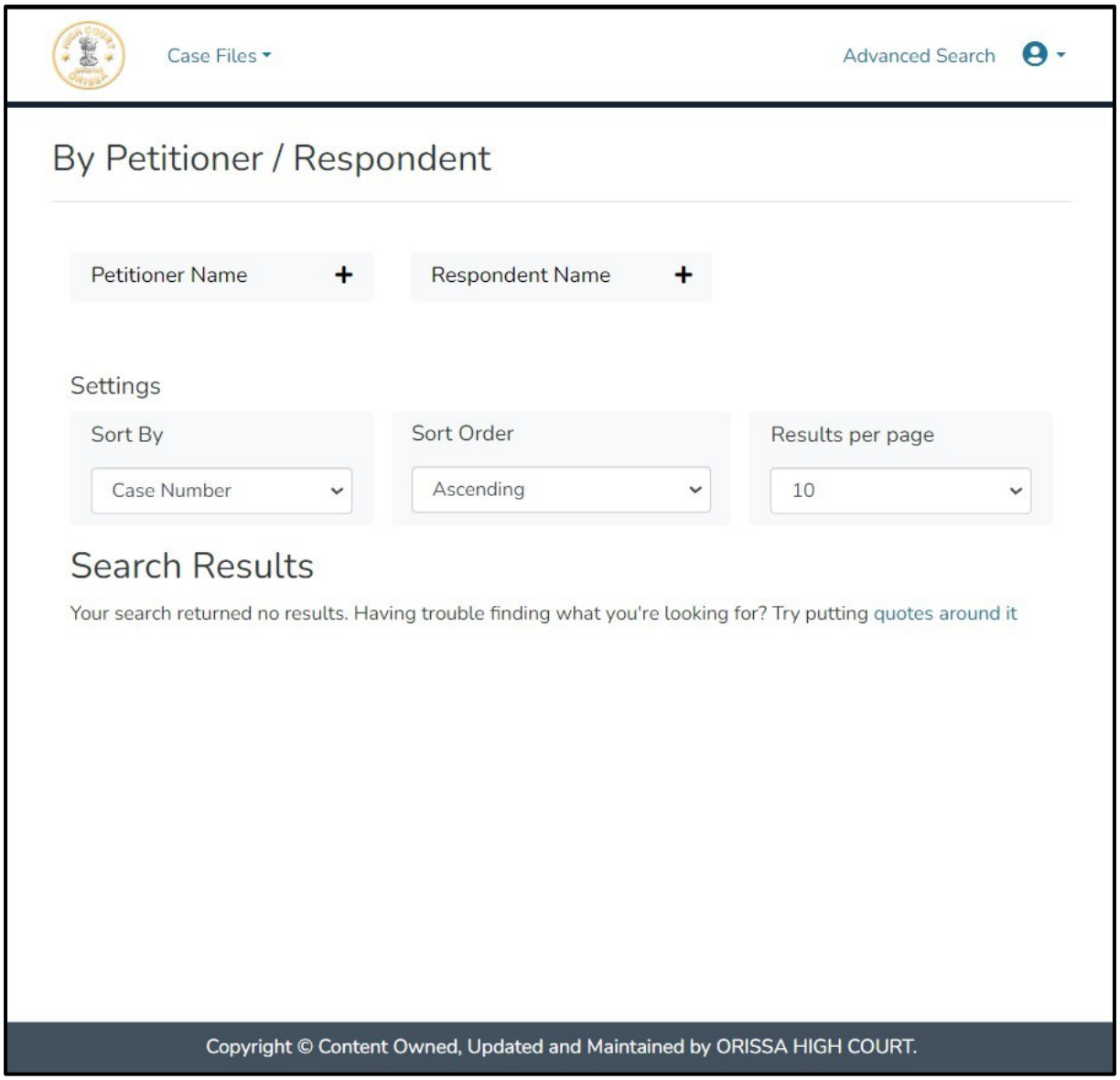
By case type / case number case year

User can search case records from their type. Following screen shows how user can search with these filters.



petitioner/respondent

In this tab user can find case records by providing petitioner or respondent name.



When a User enters data in any field, suggestions appropriate to the entered data also come up as shown in the figure below.

Case Files

Advanced Search

By Petitioner / Respondent

Petitioner Name

abant

ABANTI GIRI AND ANOTHER (1)

Respondent Name

Respondent Name

Settings

Sort By

Case Number

Sort Order

Ascending

Results per page

10

Search Results

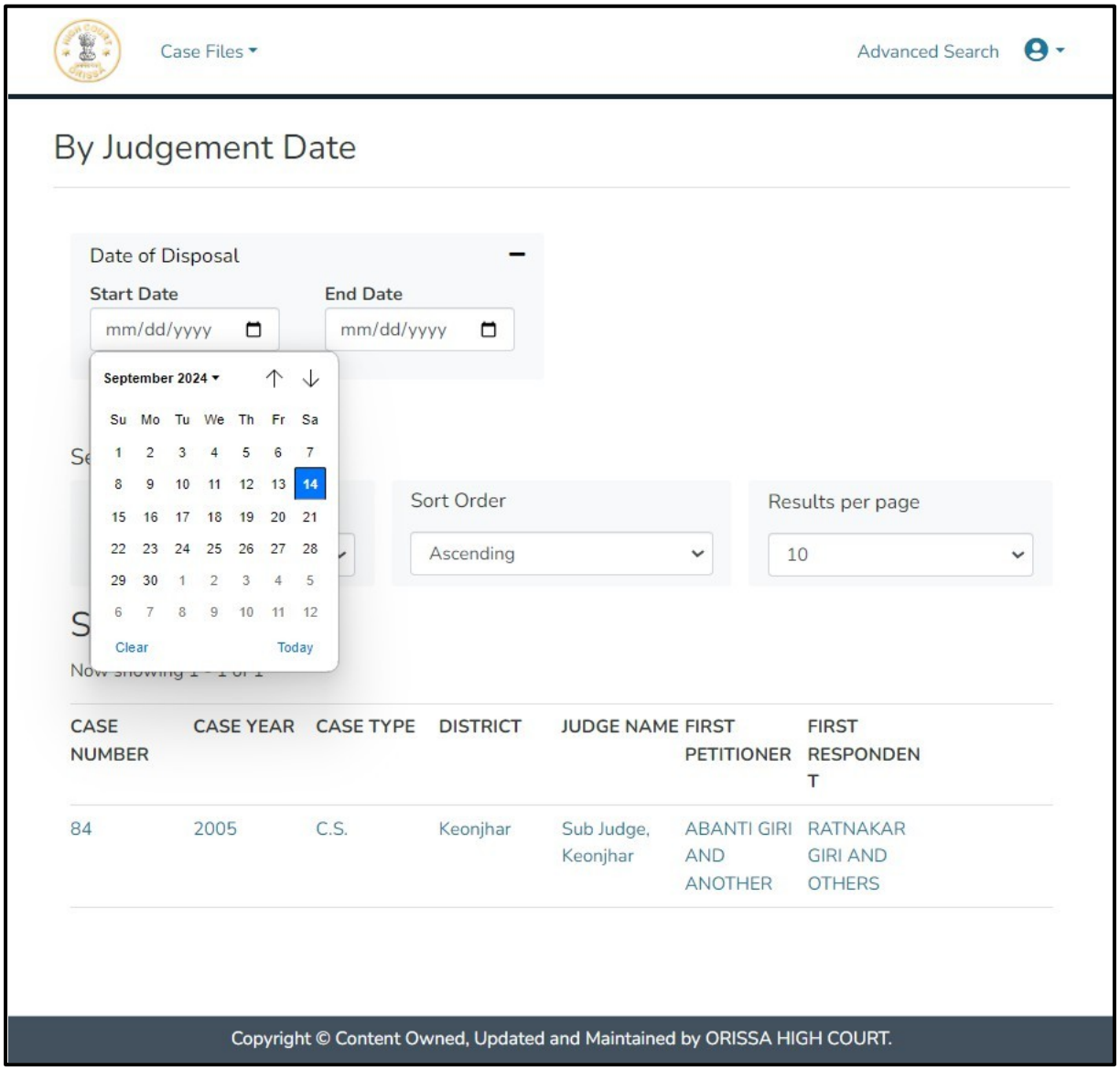
Now showing 1 - 1 of 1

CASE NUMBER	CASE YEAR	CASE TYPE	DISTRICT	JUDGE NAME	FIRST PETITIONER	FIRST RESPONDENT
84	2005	C.S.	Keonjhar	Sub Judge, Keonjhar	ABANTI GIRI AND ANOTHER	RATNAKAR GIRI AND OTHERS

Copyright © Content Owned, Updated and Maintained by ORISSA HIGH COURT.

By judgement date

Here user can search case records by providing case disposal date.



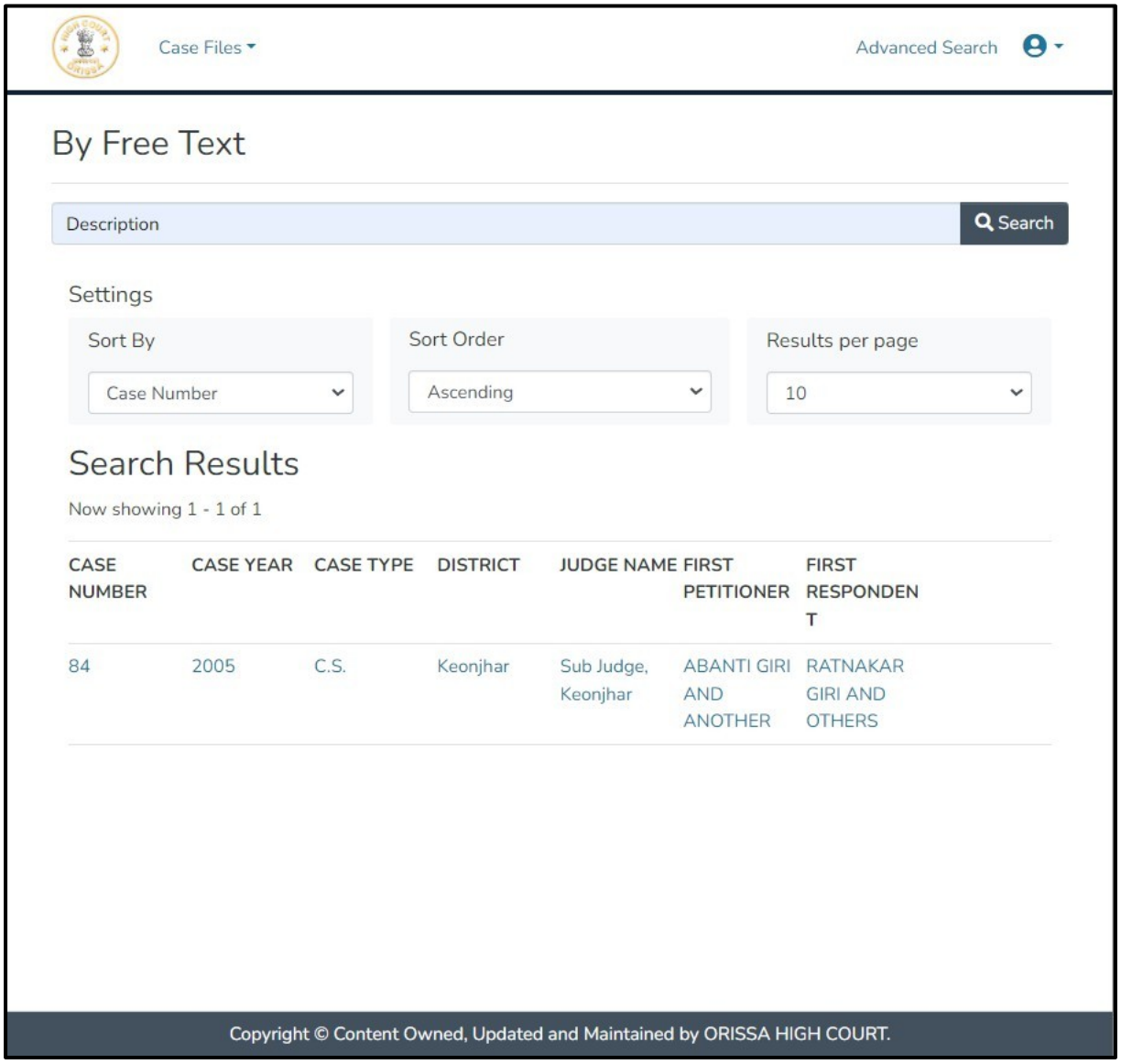
This allows users to search cases having date of decision “Equals” to a specific date or “Between” two specific dates or “After” to a specific date or “Before” to a specific date. Also uses can specify the date by using the Date Picker control as shown in the screenshot below.

By free text search

This tab searches the specified text in metadata as well as in the content of the PDF files associated with it. Here user can get case records containing the specified text.

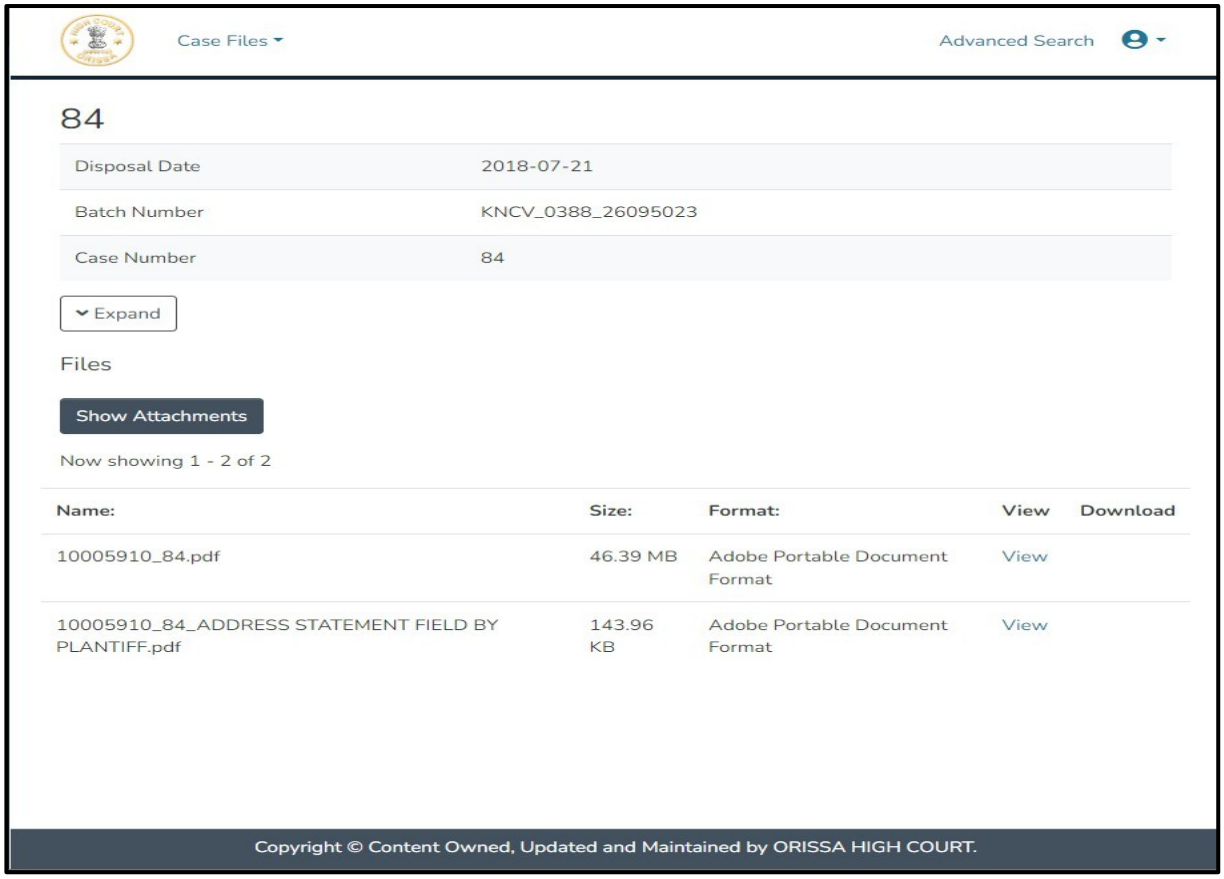
Highlighting the search words in Free Text Search

1. Searching any word in Free Text Search.

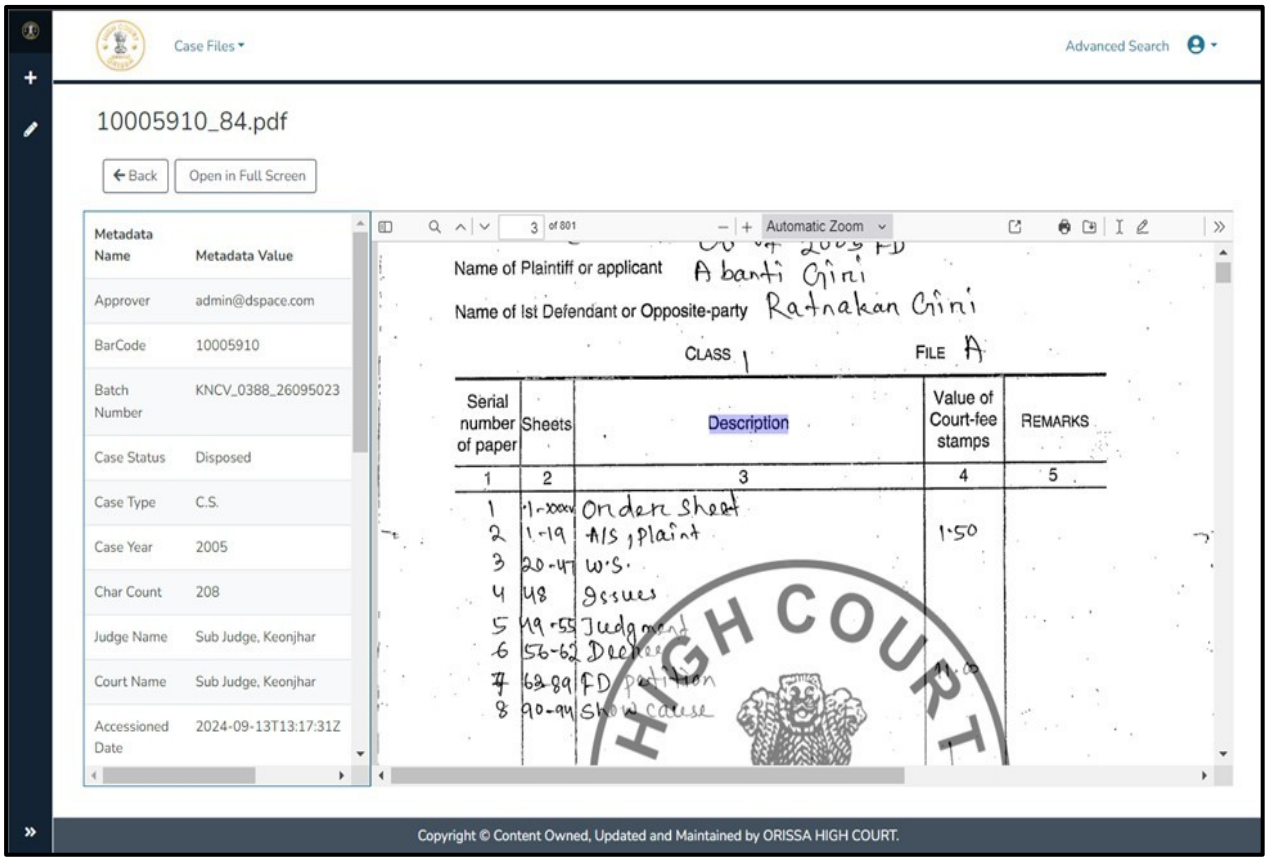


F

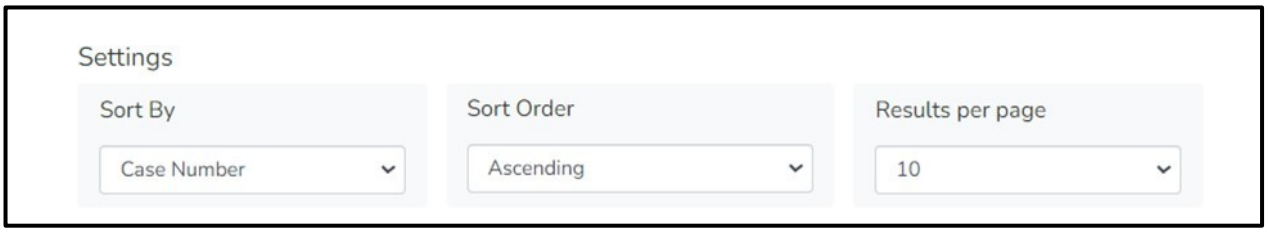
2. After opening any item, the PDF containing the search word is highlighted.



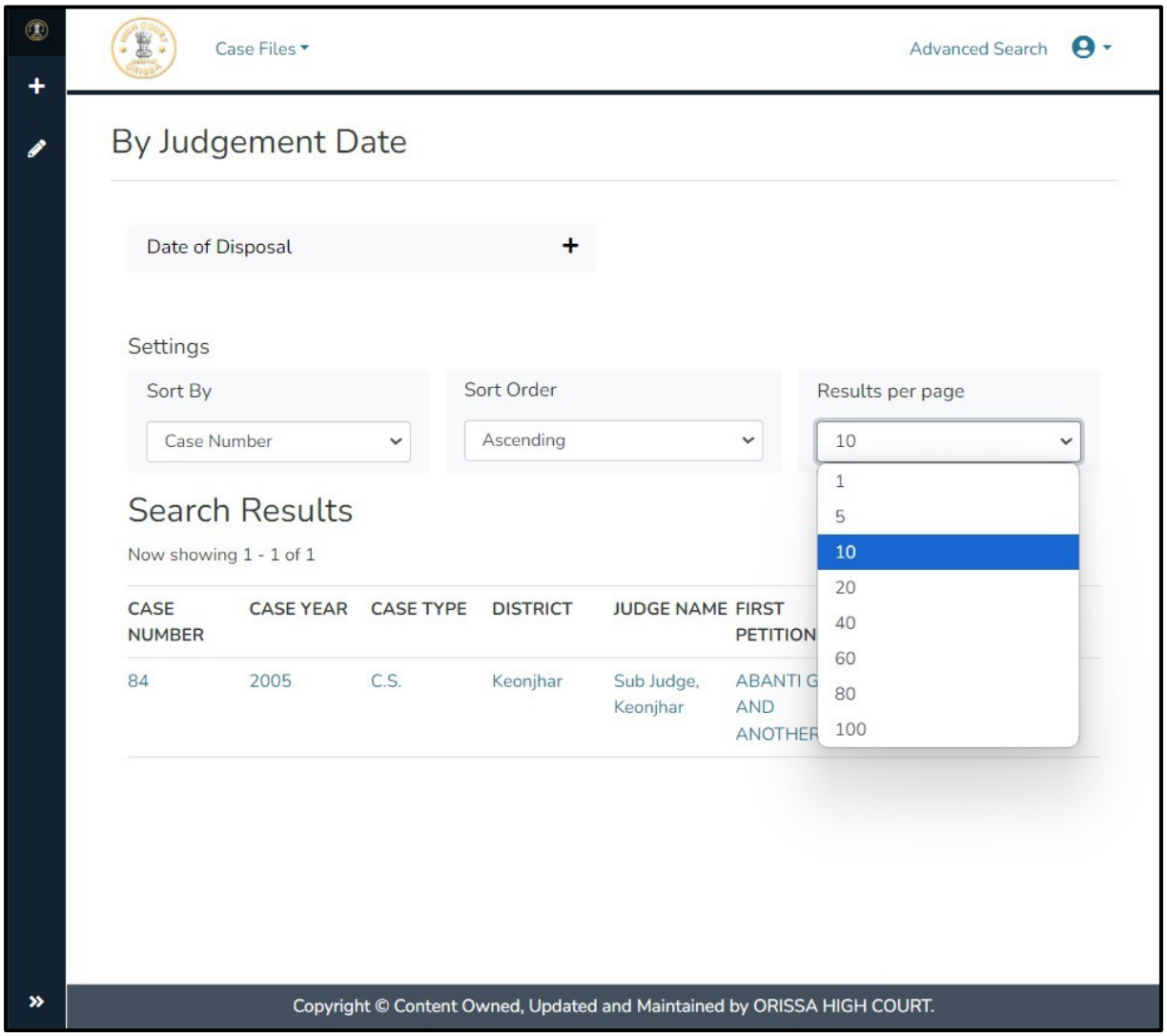
3. After opening the highlighted PDF, the search word is highlighted.



Sorting the search result

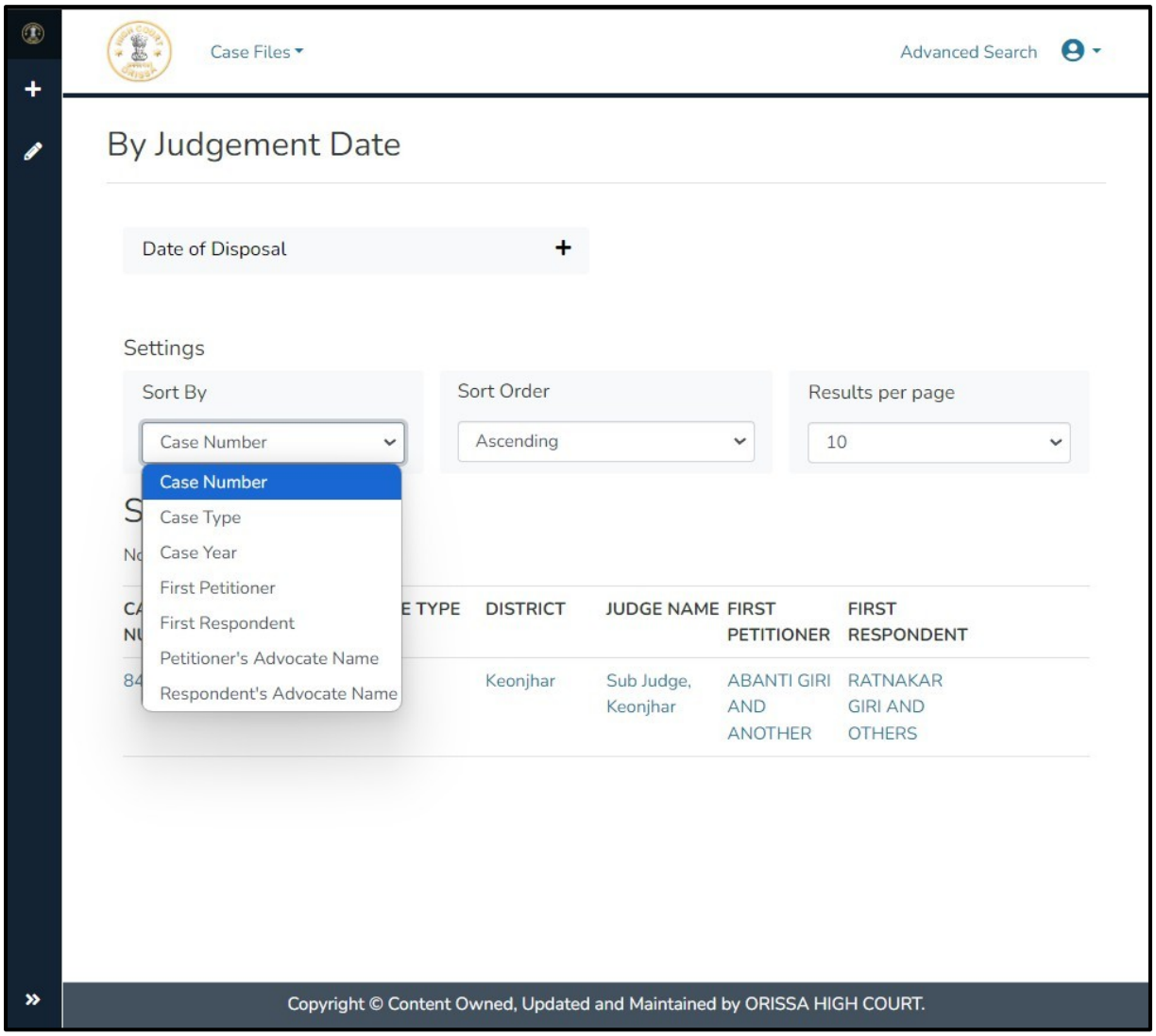


his option arranges number of case records in a page. It is a drop down list from which users can select number of results to display in one page.



sort item by

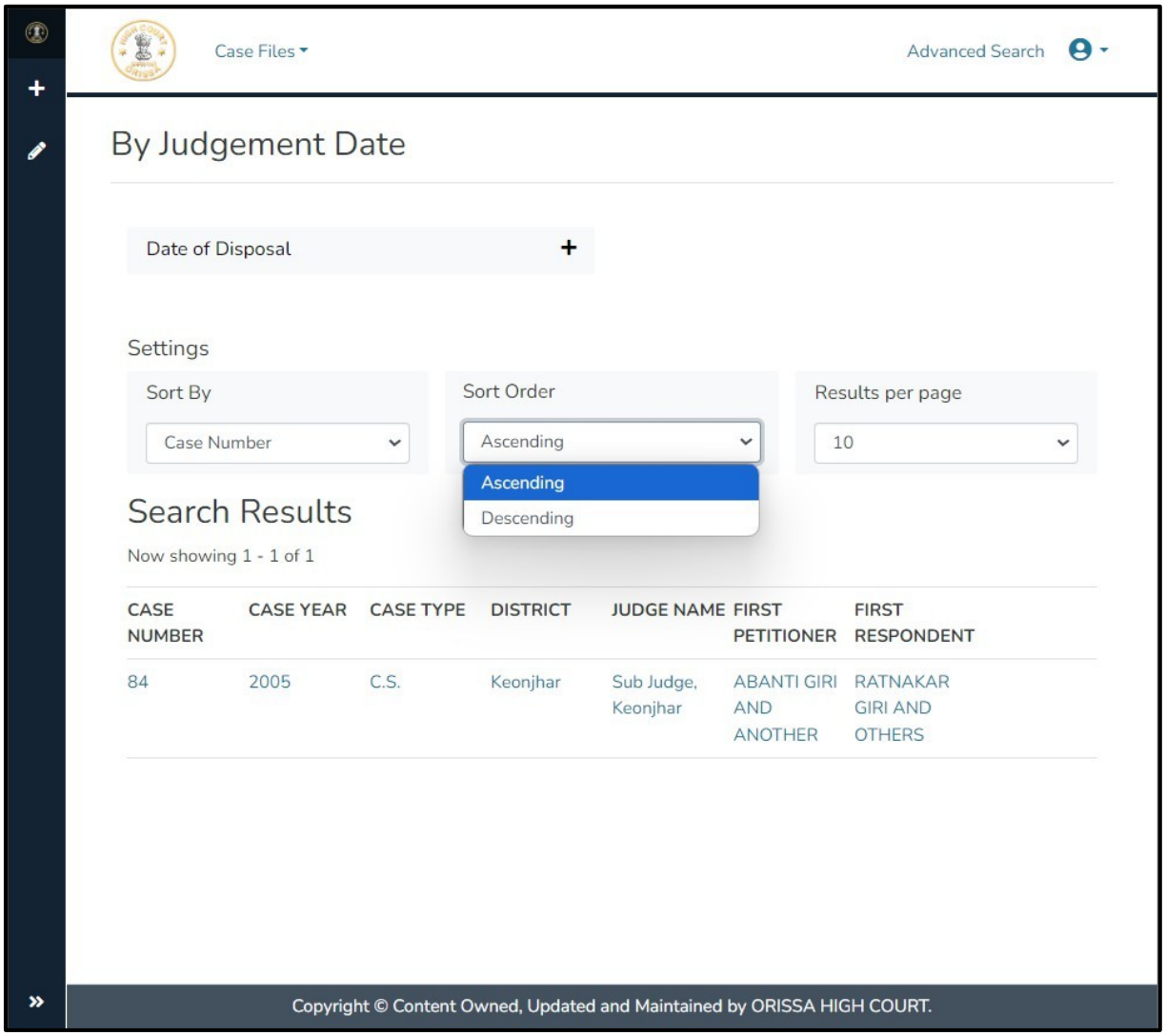
Here user can sort items based on certain conditions like “sort by relevance” to sort similar case records. It is a drop down menu from which user can select condition for sorting.



Sort order

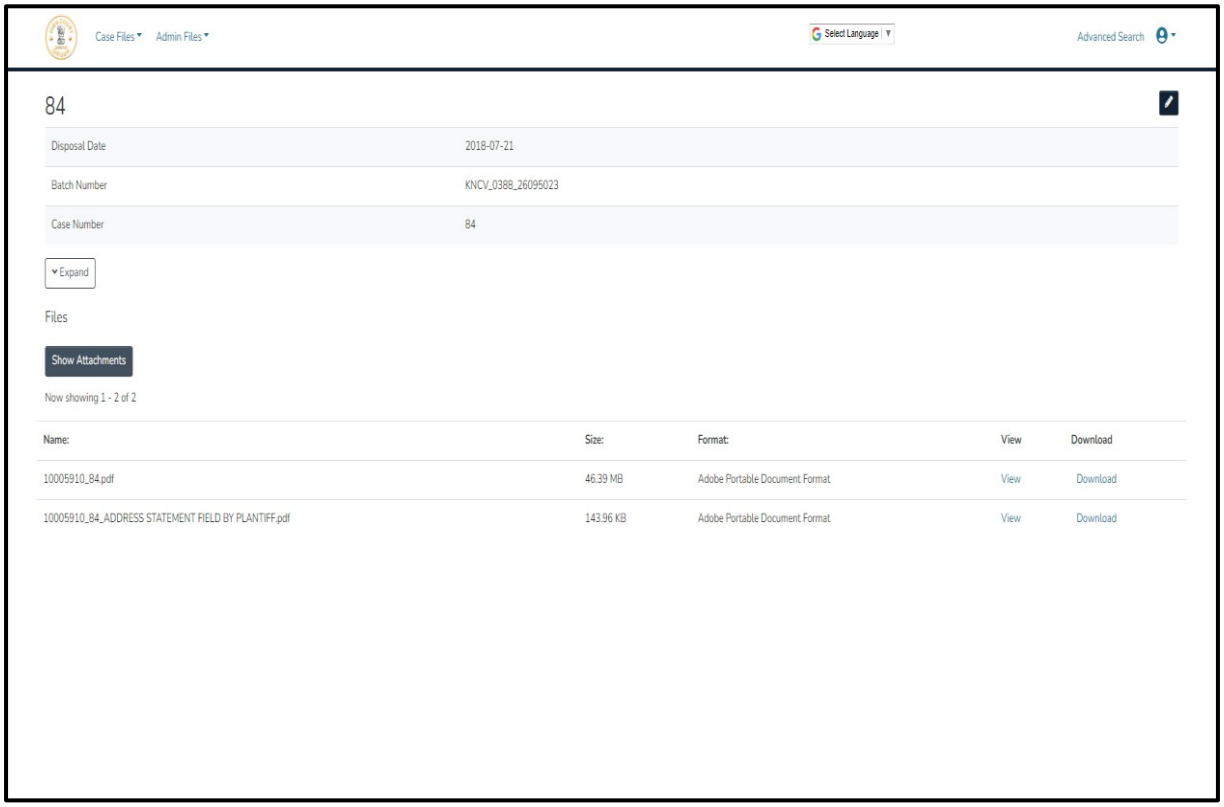
Sort order

User can sort orders in ascending or descending order of previously selected condition.



opening a case record from search resulting

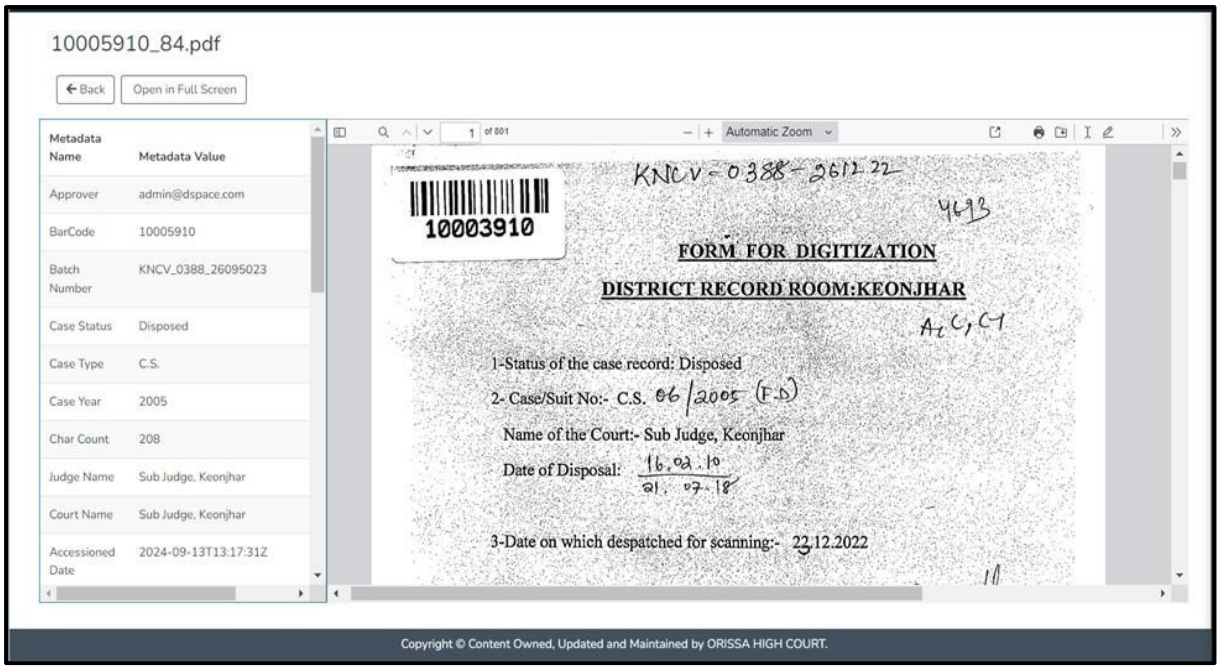
After search result is displayed user can select a case record. Following figure shows the selected search result.



Here all the information of selected record is displayed. User can open the PDF attached to the record by clicking on the view button of the PDF file or the Download button to download the PDF file.

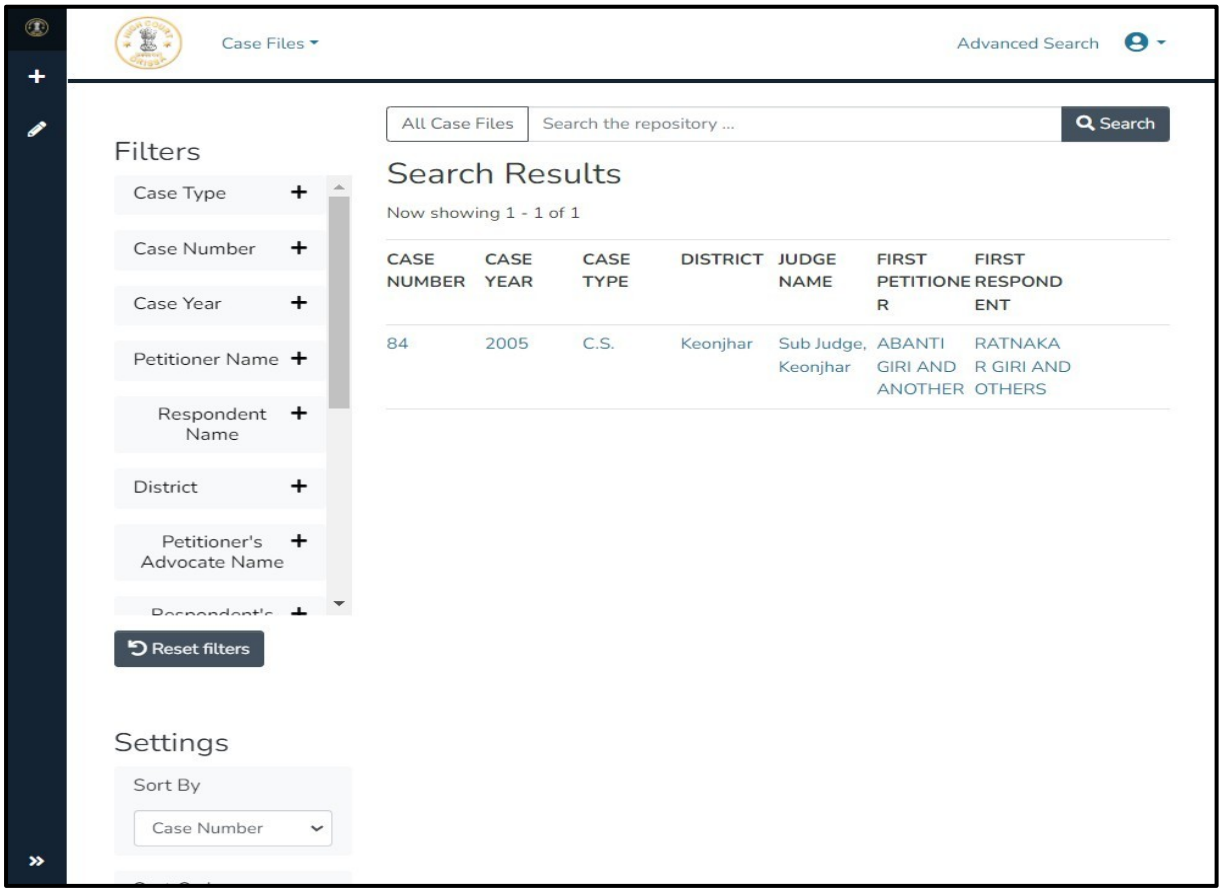
View and download pdf file

In order to view a PDF click on the PDF name link as shown above, this link is for viewing only and the original PDF is shown here without any modification but user can't download the PDFs from this link. . "Download" link allows users to download a PDF, the time of download is embedded in the bottom of the page, as shown on the image below.

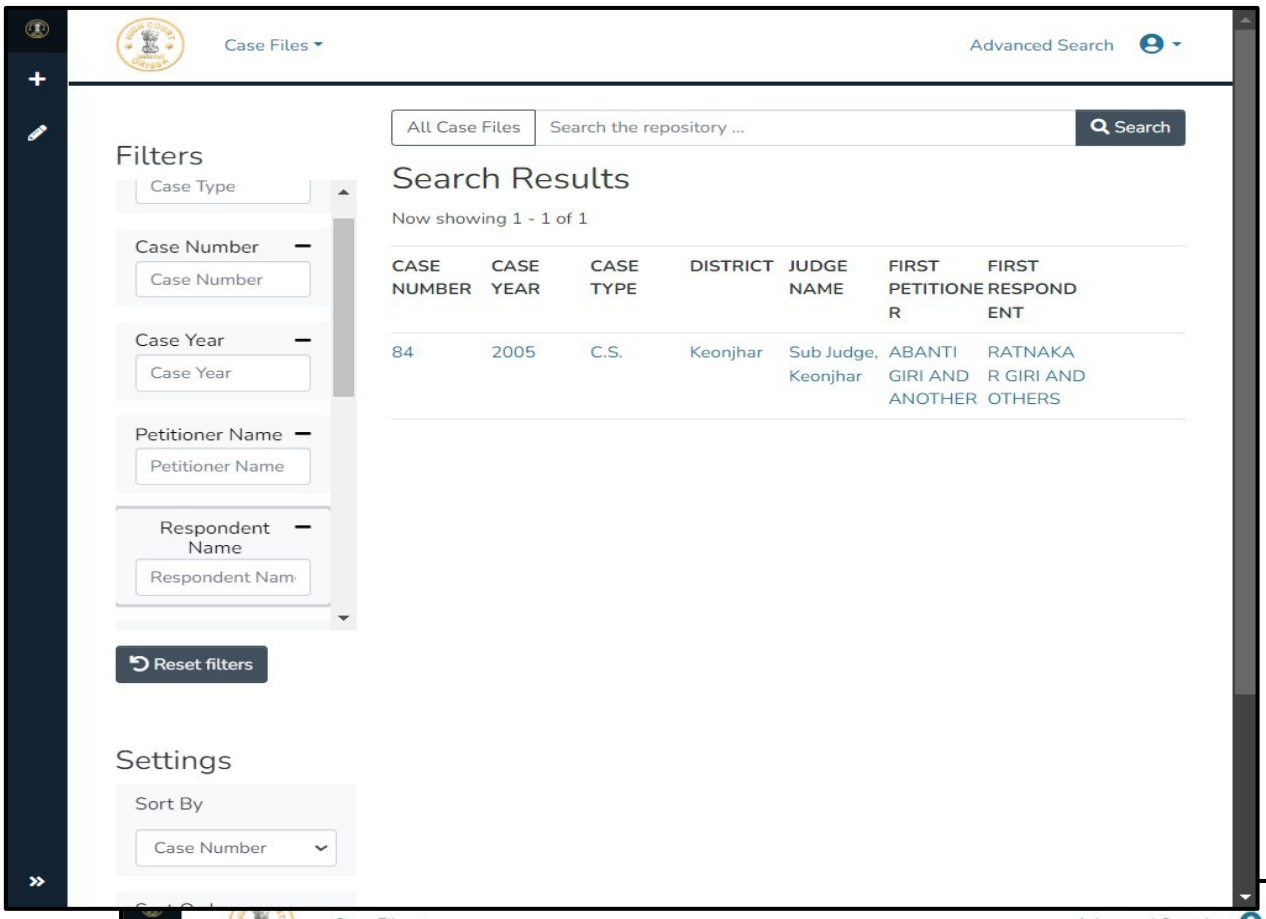


Advance search box

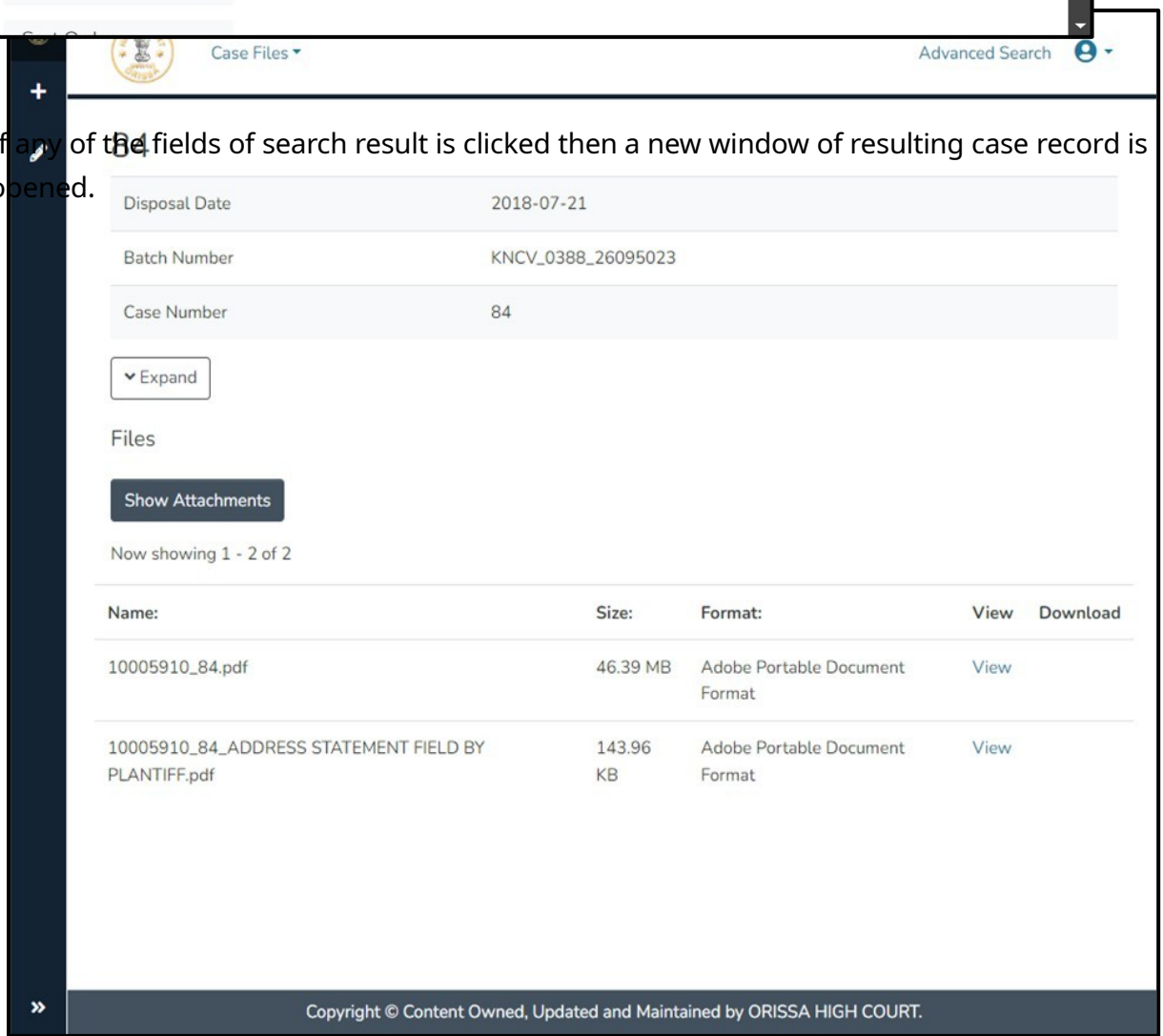
In advance search box a search page is opened as shown below.



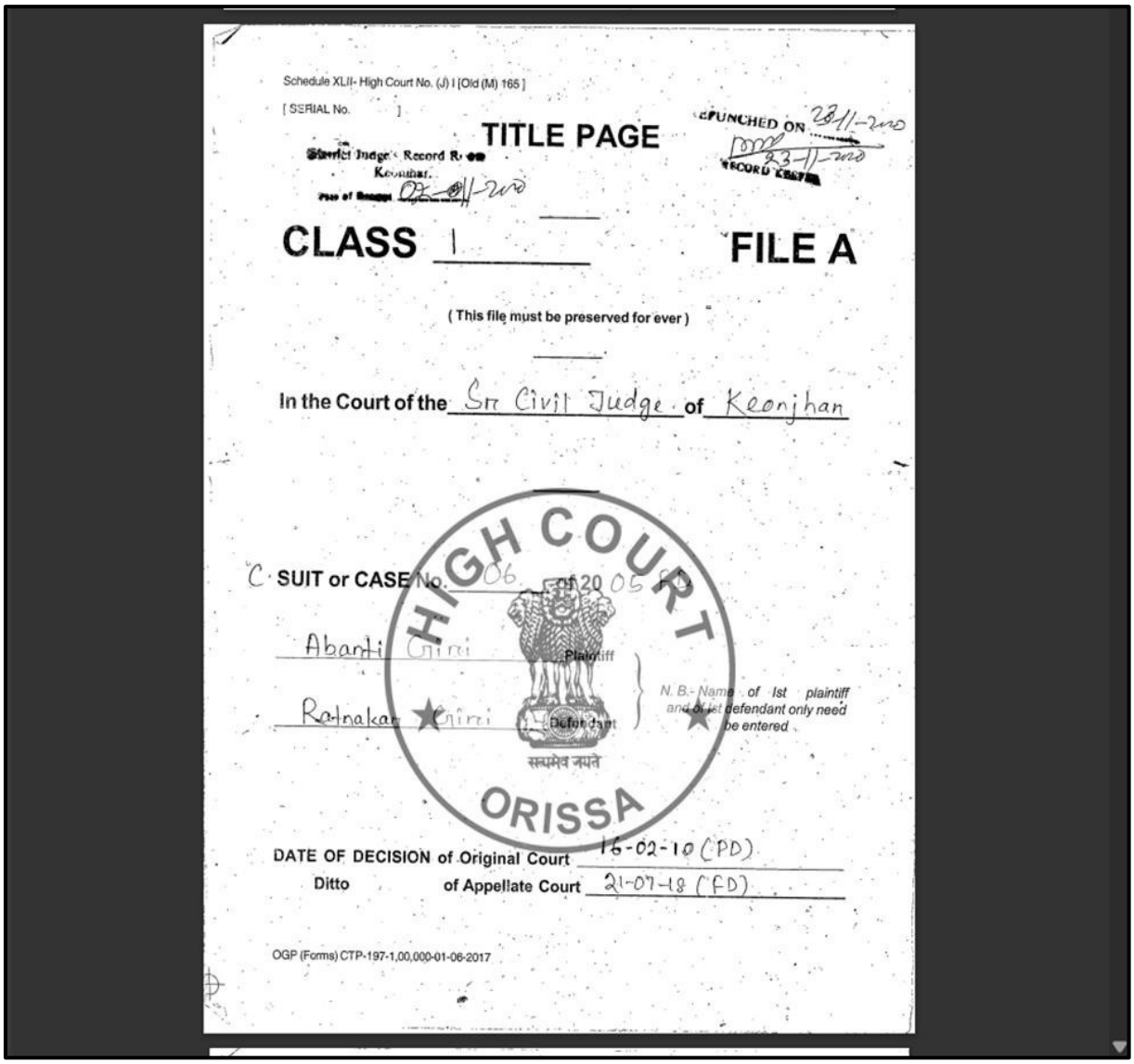
User can also add multiple filters via add button. In this way custom filters with different parameters can be applied to narrow search result.



If any of the fields of search result is clicked then a new window of resulting case record is opened.



By selecting Download Stamped if available, the PDF is opened for download with the watermark of High Court of Orissa. This is shown below.



Schedule XLII- High Court No. (J) I [Old (M) 165]

[SERIAL No.]

TITLE PAGE

PUNCHED ON 28/1-2010

RECORD KEEPER

CLASS I FILE A

(This file must be preserved for ever)

In the Court of the Sr Civil Judge of Keonjhan

SUIT or CASE No. 06 of 2006

Abanti Giri Plaintiff

Ratnakar Giri Defendant

N. B.- Name of 1st plaintiff and of 1st defendant only need be entered

HIGH COURT ORISSA

सत्यमेव जयते

DATE OF DECISION of Original Court 15-02-10 (PD)

Ditto of Appellate Court 21-07-18 (FD)

OGP (Forms) CTP-197-1,00,000-01-06-2017

adding comments for a pdf

When a PDF is chosen for viewing, user has an option for adding comments to the document.

When a comment is added, a pop comes confirming the success of the action. Subsequently the comment is added to the PDF with additional information of the author of the comment and the time and date of the comment. This is shown below.

Case Files Admin Files Select Language Advanced Search

10005910_84.pdf

Back Open in Full Screen

Metadata Name	Metadata Val
Approver	admin@dspac
BarCode	10005910
Batch Number	KNCV_0388_
Case Status	Disposed
Case Type	C.S.
Case Year	2005
Char Count	208
Judge Name	Sub Judge, Ke
Court Name	Sub Judge, Ke
Accessioned Date	2024-09-13T
Available Date	2024-09-13T

1 of 801 Page

10003910

FORM FOR DIGITIZATION

DISTRICT RECORD ROOM:KEONJHAR

1-Status of the case record; Disposed

2- Case/Suit No:- C.S. 96/2005

Name of the Court:- Sub Judge, Keonjhar

Date of Disposal:- 15.04.19

3-Date on which despatched for scanning:- 23.12.2022

Signature of the concerned employee

4-Date on which scanned

Signature of the scanning officer

5-Date of verification

Signature of the verifying officer

6-Date on which re-scanned

Signature of the concerned employee

HIGH COURT ORISSA

Export

Add a comment Add Comment

Copyright © Content Owned, Updated and Maintained by ORISSA HIGH COURT.

Case Files Admin Files Select Language Advanced Search

10005910_84.pdf

Back Open in Full Screen

Metadata Name	Metadata Val
Approver	admin@dspac
BarCode	10005910
Batch Number	KNCV_0388_
Case Status	Disposed
Case Type	C.S.
Case Year	2005
Char Count	208
Judge Name	Sub Judge, Ke
Court Name	Sub Judge, Ke
Accessioned Date	2024-09-13T
Available Date	2024-09-13T

1 of 801

10003910

FORM FOR DIGITIZATION

DISTRICT RECORD ROOM:KEONJHAR

1-Status of the case record; Disposed

2- Case/Suit No:- C.S. 96/2005

Name of the Court:- Sub Judge, Keonjhar

Date of Disposal:- 15.04.19

3-Date on which despatched for scanning:- 23.12.2022

Signature of the concerned employee

4-Date on which scanned

Signature of the scanning officer

5-Date of verification

Signature of the verifying officer

6-Date on which re-scanned

Signature of the concerned employee

HIGH COURT ORISSA

Export

admin@dspace.com - 2024-09-16 11:05

Comment - This case is for petitioner

Delete

This case is for petitioner Add Comment

Copyright © Content Owned, Updated and Maintained by ORISSA HIGH COURT.

The comments can be removed by clicking on the trash icon. A confirmation page comes to reconfirm the action. After confirming a popup comes confirming the success of the action.

Step 2: Metadata Creation & File Upload

1. Metadata Creation

- Define and structure metadata based on court records.
- Ensure compatibility with judicial archiving standards. We will provide all the metadata as per the court record.

2.File Upload Mechanism

- Develop an uploader to handle single and bulk file uploads.
- Support zip format for bulk upload.
- Upload all types of court records, including audio, video, text, and other electronic formats.
- Automatically convert uploaded PDFs to PDF/A format.

Step 3: Case Record Verification Module

- After a case record is uploaded, it will be dsplayed in the **admin user pool**.
- The **admin user** assigns the record to specific individuals or groups.
- Assigned users will **verify** the records.
- If verified, the record is **approved** and becomes visible to others.
- If rejected, the record is sent back to the **admin user**.
- A **comment section** will be available for users to provide reasons for rejection.

Search & Retrieval Customization

The access portal will support multiple search mechanisms for retrieving digital records efficiently:

1. Query in Multiple Languages

- Enable search functionality in **English, Hindi, and regional languages**.
- Example: A user searching for "land dispute" in Hindi (जमीनी विवाद) should receive relevant results.

2. Fuzzy Search

- Allows approximate matches to account for spelling errors or variations.
- Example: Searching for "rajesh jee" may still return results for "rajesh jee".

3. Full-Text Search

- Enables users to search within document content.
- Example: A lawyer searching for "contract breach clause" retrieves documents containing these words in their text.

4. Cataloging Metadata Search

- Structured search based on predefined metadata fields.
- Example: Searching by case number ("2024-CIV-5678") retrieves the exact case file.

5. Faceted Search with Filtering Mechanisms

- Allows users to filter results based on different categories.
- Example: A search for "property case" can be refined by **year, jurisdiction, case type, or document type**.

6. Boolean Search Mechanism

- Supports logical operators (**AND, OR, NOT**) to refine searches.
- Example: Searching "land AND dispute" will return only documents containing both words.

7. Wildcard and Proximity Search

- Wildcard searches use "*" or "?" to replace unknown characters.
- Proximity search finds words within a specific distance.
- Example: Searching for "law*" will return "lawyer, lawsuit, laws".
- Example: "contract breach"~5 finds these words within 5 words of each other.

8. Nested Keyword Search (Search within Search)

- Enables searching within an already filtered result set.
- Example: Searching for "contract" first, then filtering for "2022" within those results.

9. Elastic Search for Enhanced Performance

- Uses indexing for high-speed retrieval and ranking.
- Example: Searching for "civil rights" will return the most relevant documents ranked by relevance and frequency.

Step 4: Storage & Digital Document Preservation

- Support for long-term preservation of digital documents, including court records, audio-video material, and other electronic formats.
- Ensure that all uploaded PDFs are secured using **256-bit AES encryption**.
- Automatically convert any user-uploaded PDF to **PDF/A format** within the DMS for compliance with archival standards.
- All files should display a watermark, and the file download option should also include a watermark that varies based on the client.

Step 5: Digital Signature for Case Records

- Enable **digital signing** of case records.
- Ensure signed copies are served as **certified copies**.
- Implement user-based tracking for digital signatures.
- Court judges will use **dongles** for enabling **digital signatures**.
- Digital signatures will only work with **physical DSC Token** to ensure security.

Step 6: Data Harvesting Configuration

- Configure **DSpace's default data harvesting capabilities** as per judicial requirements.
- Ensure structured data collection, indexing, and storage.

Step 7: User & Group Management (Modified DMS User System)

- Modify the default DSpace user management system to align with court-specific requirements.
- Implement a hierarchical user structure:
    - **Administrator**: Holds full access and control over all actions.
    - **User Groups**: Admin can create and manage multiple user groups with varying permission levels.

Access Restrictions & Permissions

- Admin can restrict specific user groups from viewing certain case file items, such as **order sheets**.
- Implement **collection-based access control**, allowing the admin to grant or revoke access to specific document sets.
- Support **time-based access**, enabling the admin to assign temporary access rights to users or groups.
- Users must view documents using the **internal document viewer** of the website.
- Any downloaded PDF will be **password-protected**, with each user assigned a unique password for tracking.
- Embed user details within downloaded PDFs to identify the individual who accessed the document.
- Ensure seamless role management through an intuitive **UI-based control panel** for administrators.

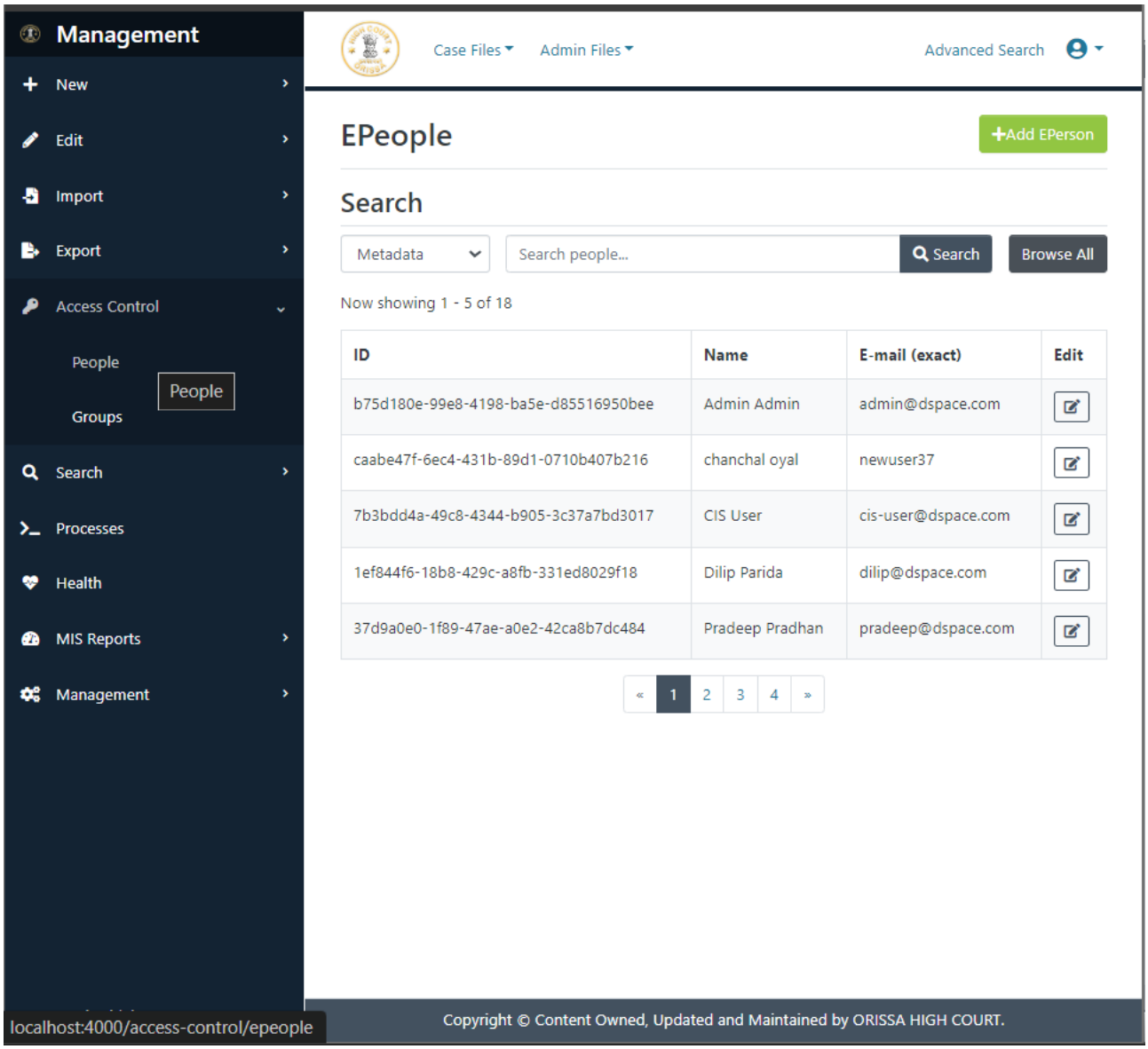
Note: Images will be shared for better clarity regarding the existing DMS user management structure and planned modifications.

E-people

Under this tab administrator can add users, edit user details and delete user.

Add E- Person adds a user. For editing and deleting a user, user must be selected first by using select

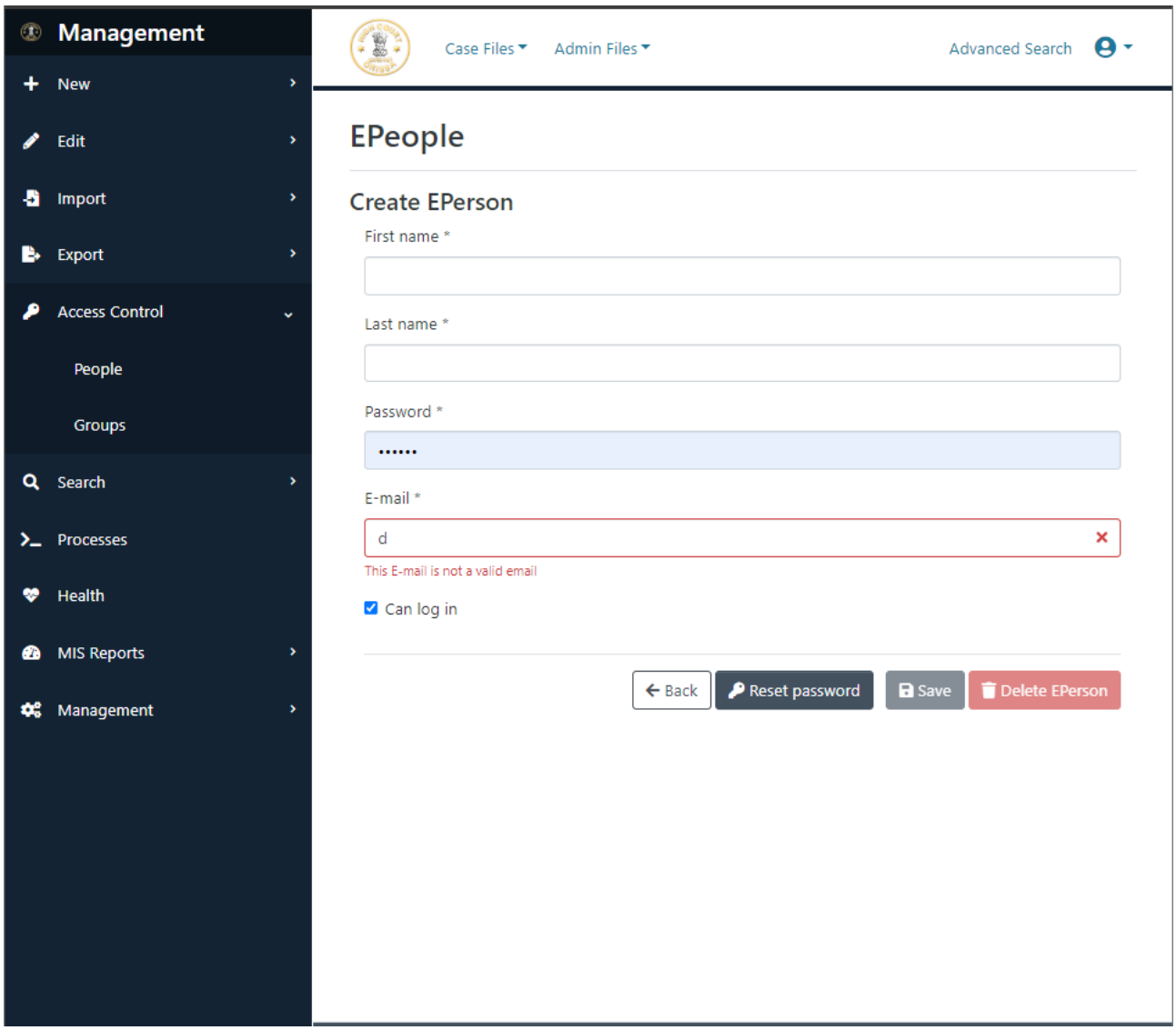
E- Person button.



On saving details, user is created and control navigates back to E-People page. Administrator can verify user by using select E-Person button.

To modify user, administrator has to select E-Person. On clicking E-Person a page pops up displaying

list of users.



When administrator select a particular user the text box above select E-Person button displays the name of that user. Now administrator can use edit or delete options provided next to the E-Person button.

On clicking edit button details of that user appear, which could be edited. On clicking delete button user is removed.

Case Files Admin Files Advanced Search

EPeople

Edit EPerson

First name *

chanchal

Last name *

oyal

Password *

E-mail *

newuser37

Must be valid e-mail address

☑ Can log in

Back | Reset password | Save | Delete EPerson

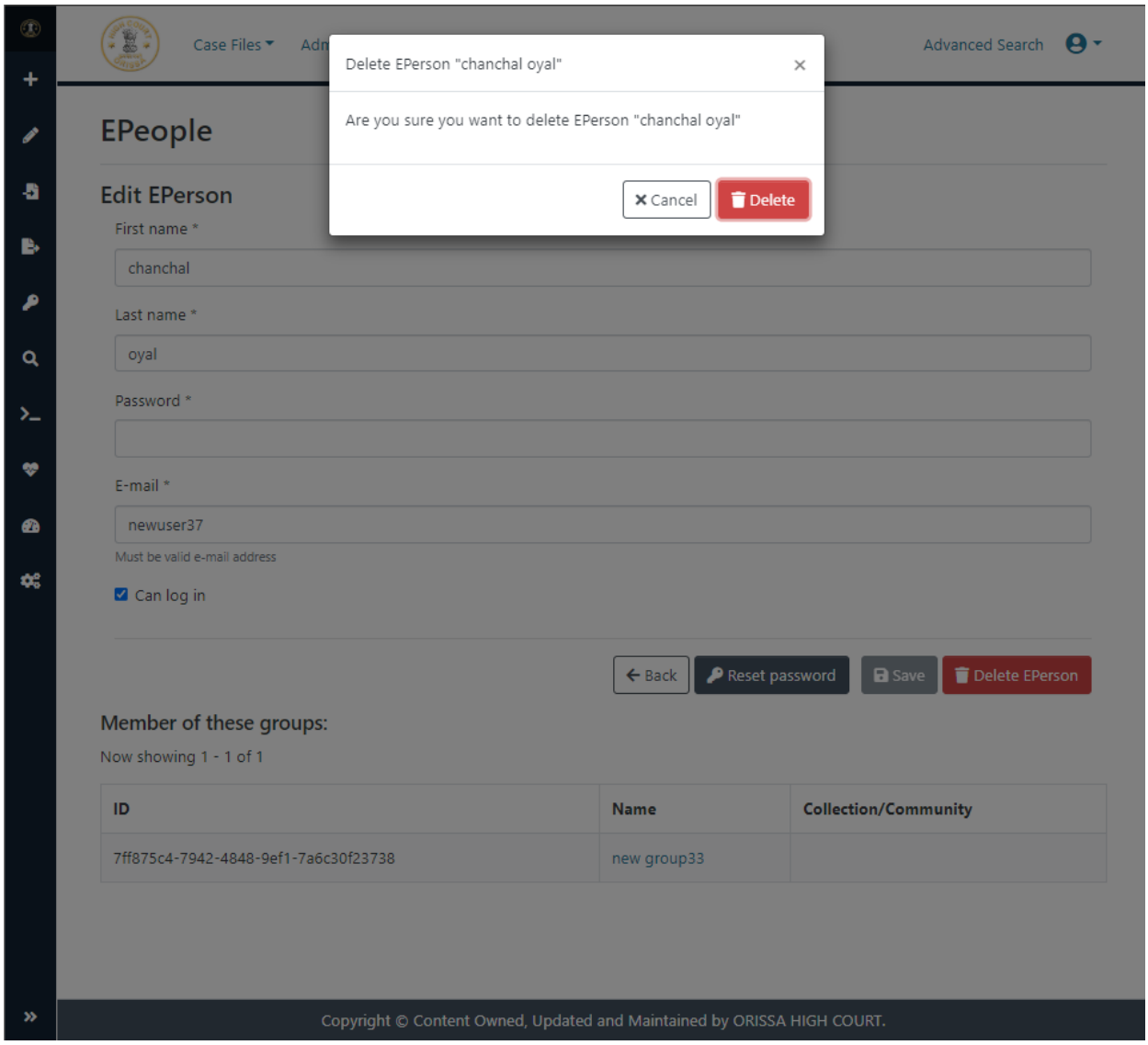
Member of these groups:

Now showing 1 - 1 of 1

ID	Name	Collection/Community
7ff875c4-7942-4848-9ef1-7a6c30f23738	new group33	

Copyright © Content Owned, Updated and Maintained by ORISSA HIGH COURT.

Before deleting a user a warning is displayed to prevent accidental deletion.

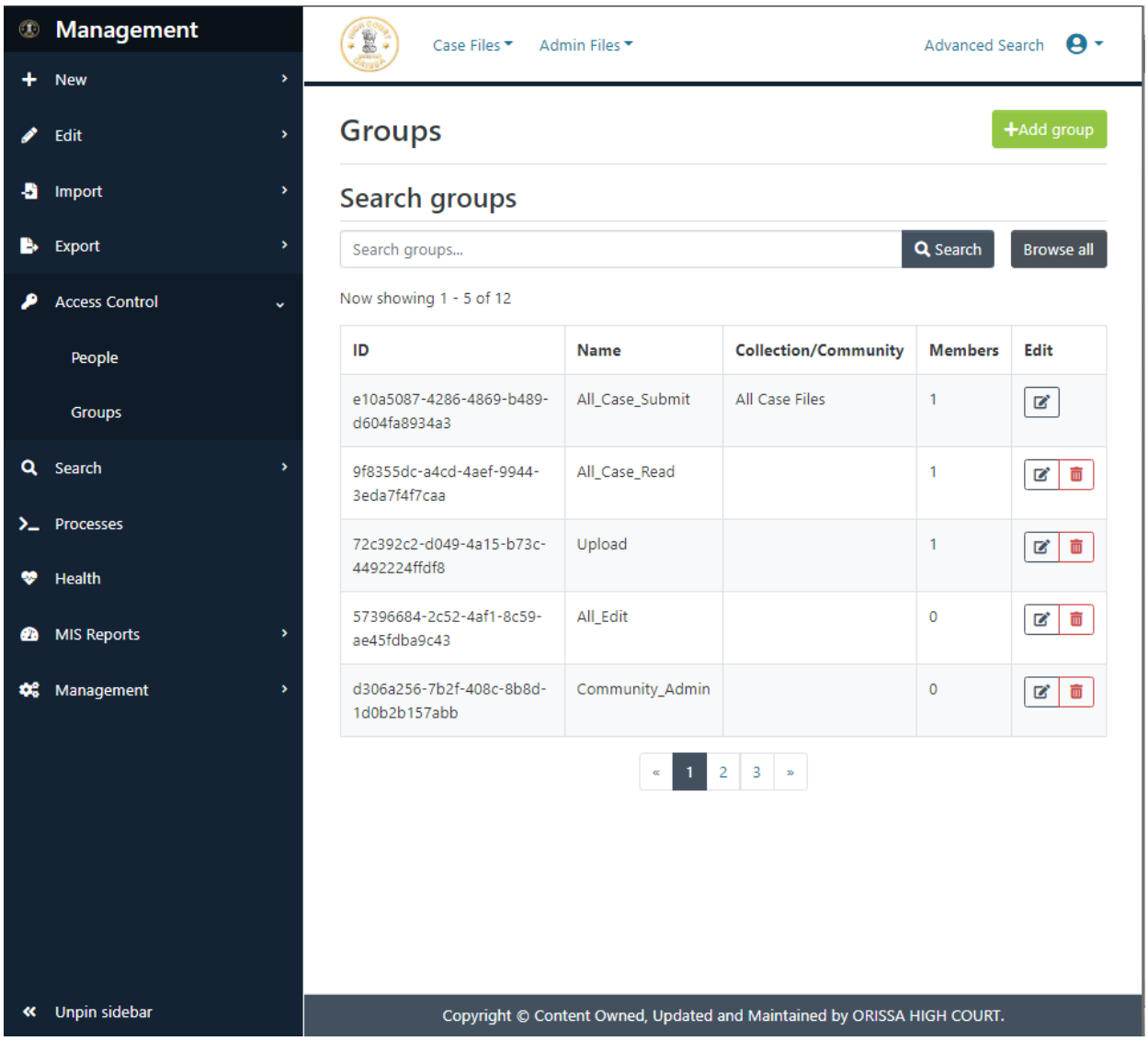


Groups

When create new group is clicked then a new page asking group information is opened. This page is

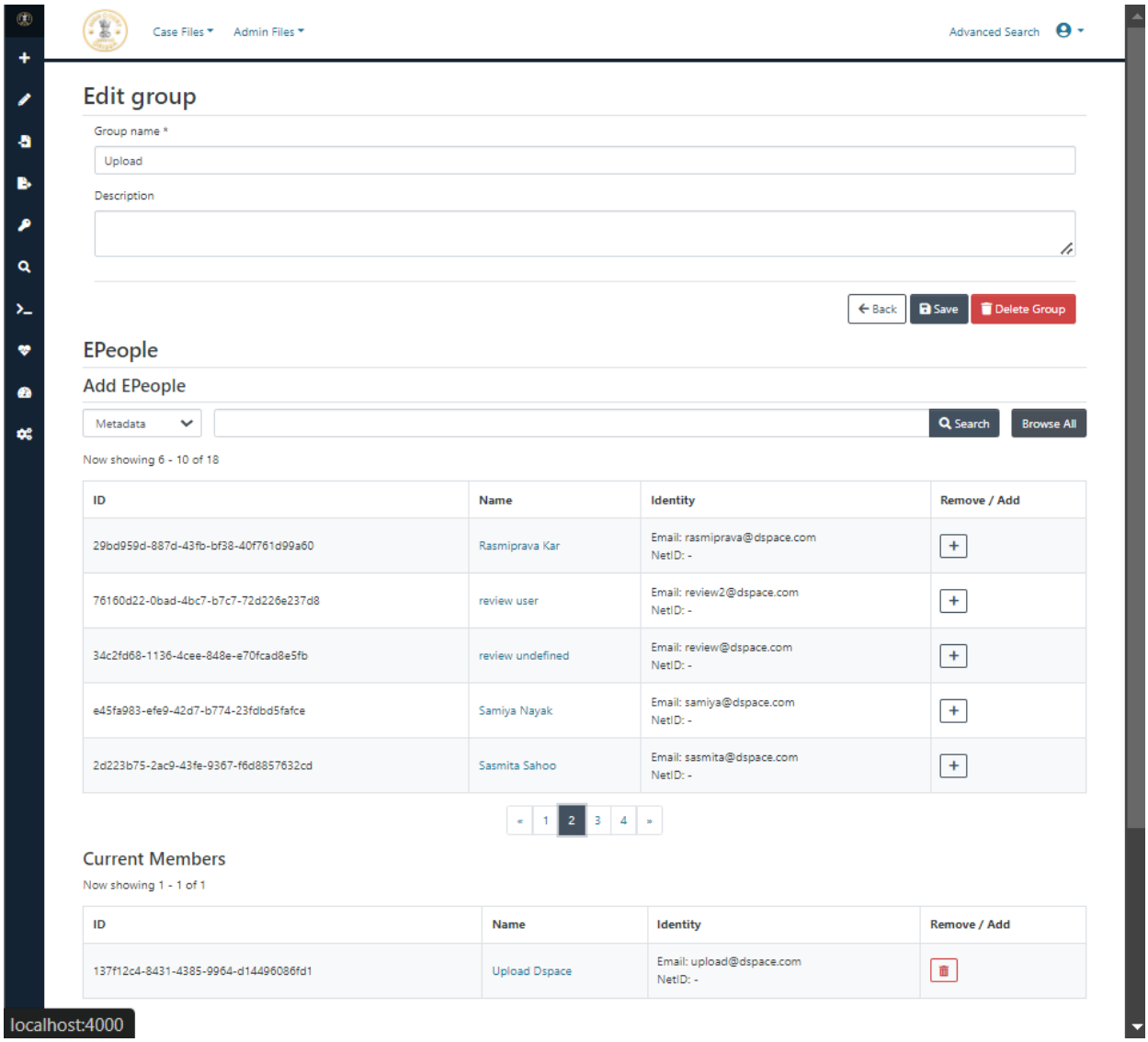
same in edit option under current menu. After editing/adding user information administrator has to

click update button to save the information.



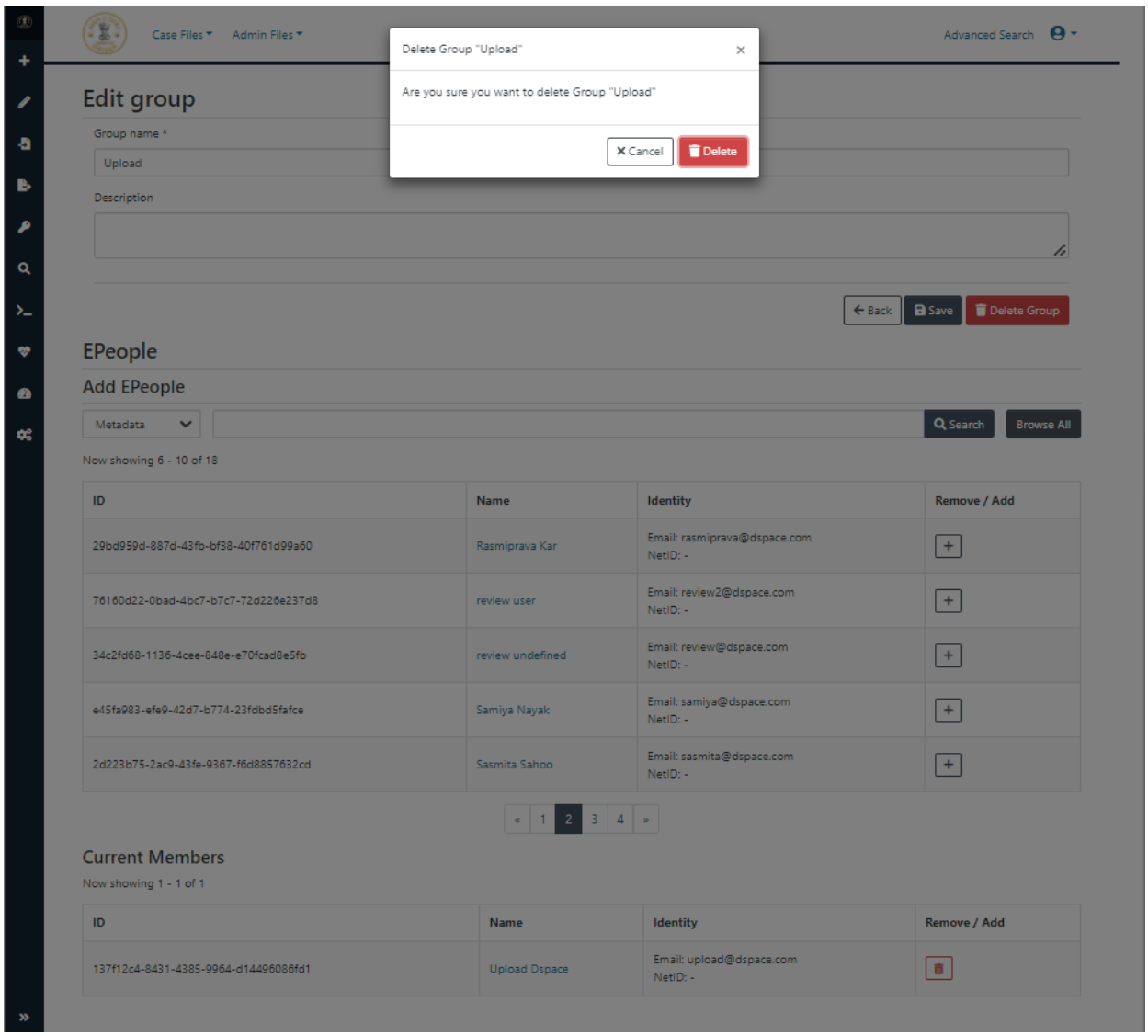
Epersons can be added by selection one from the Select E-people list

The admin user needs to save any changes made to the group by clicking Update Group.



Administrator can also delete a group via delete button. Before deletion a warning is issued to

prevent accidental deletion.



Predefinded groups

˅

All_Case_Read

Member of this group have complete read permission on the collection All Case Files.

˅

All_Case_Submit

Member of this group have complete write and submission permission on the collection All Case

Files.

˅

Upload

Member of this group have an option in the home page for importing batch.

˅

ReadOnly

Member of this group will not be given an option to download a PDF.

˅

Community_AdminMember of this group have complete admin permission on the collection All Case Files.

˅

Administrator

Member of this group have complete admin permission on all collection.

˅

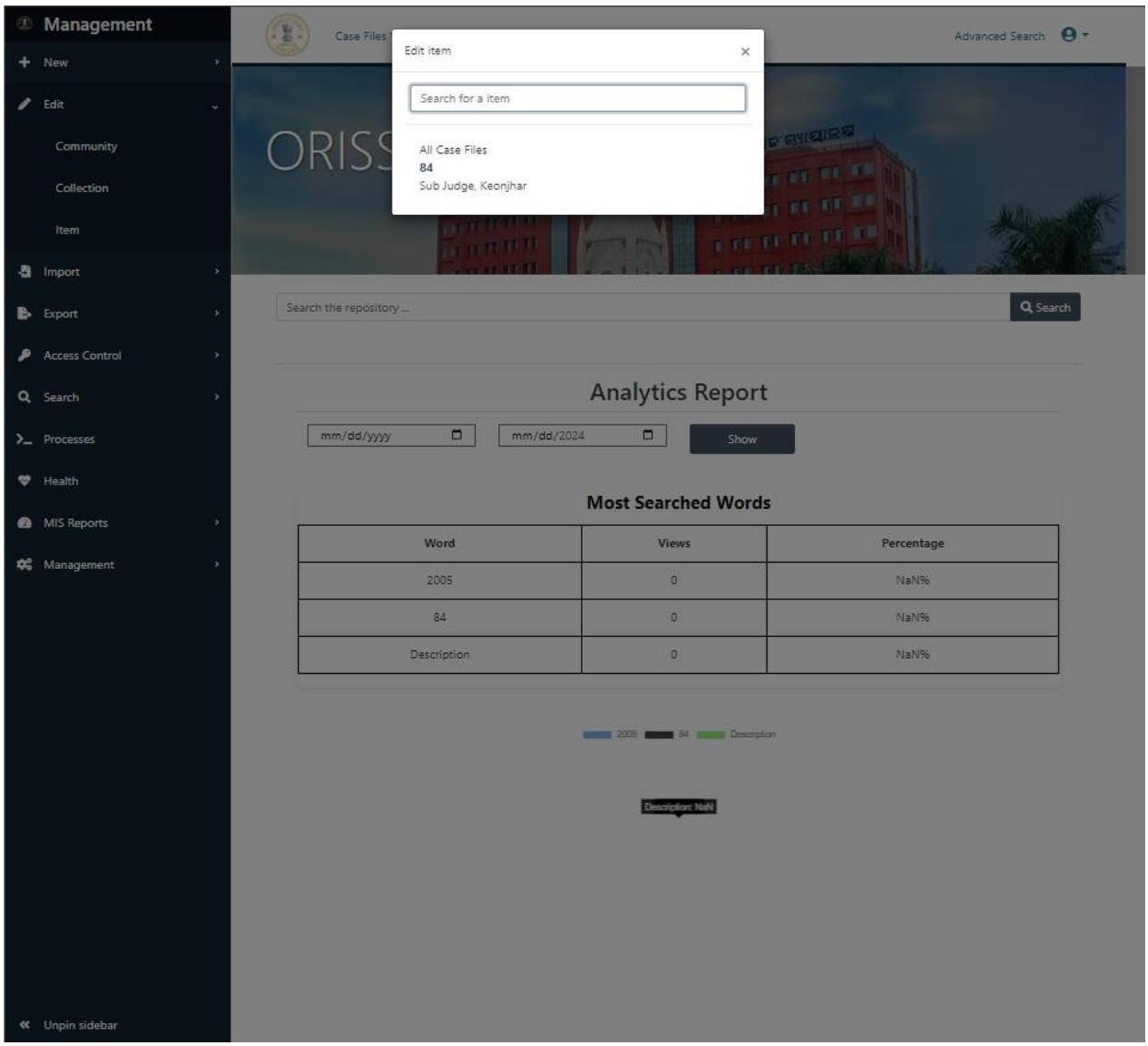
Anonymous

Member of this group will be treated as if they have not logged in.

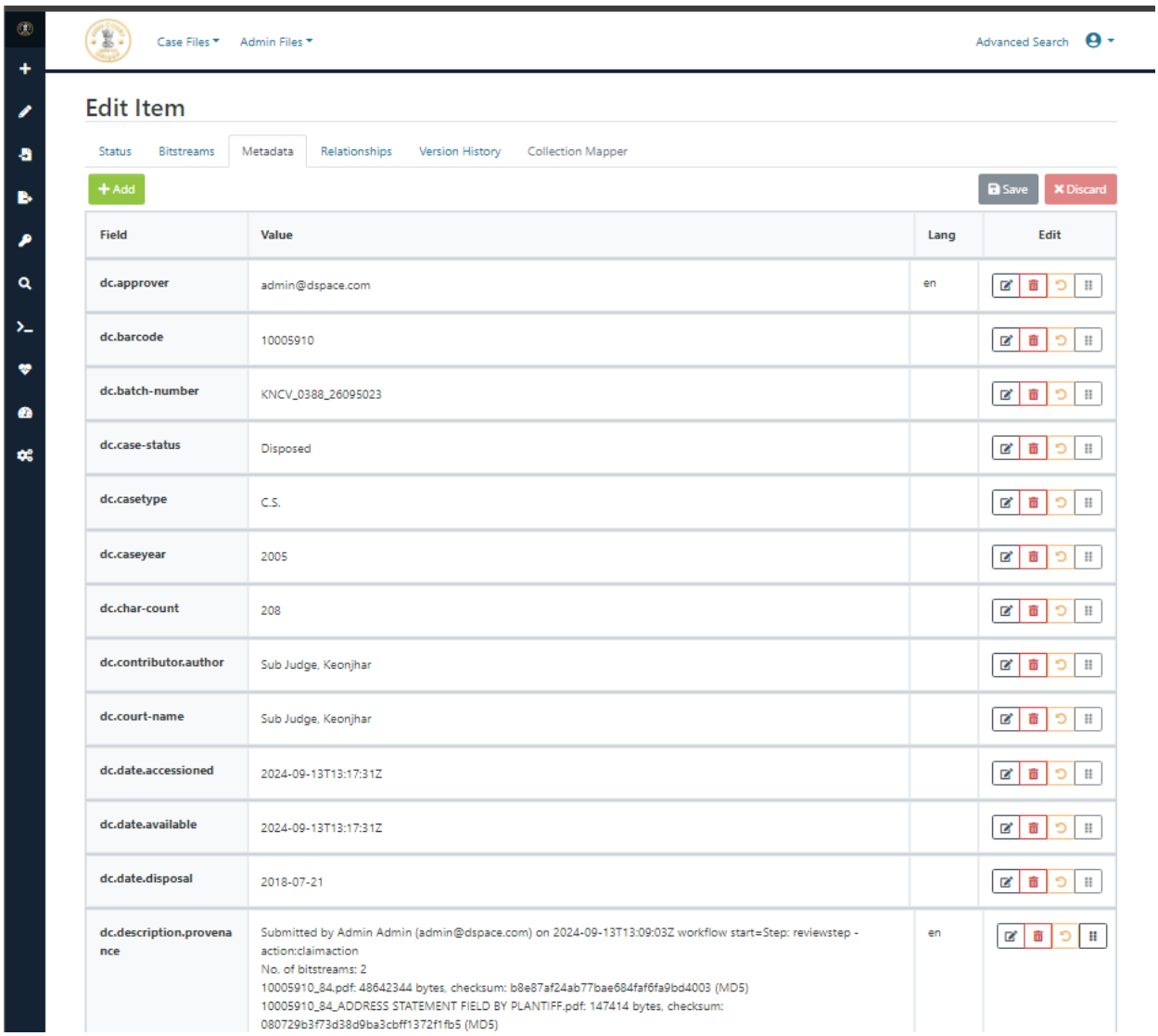
Edit items

Here administrator can edit or delete an item (case record). First page displays a search box from

which a particular item can be found by providing items handle number or internal id.



After pressing find button, item’s metadata record is displayed as follow.

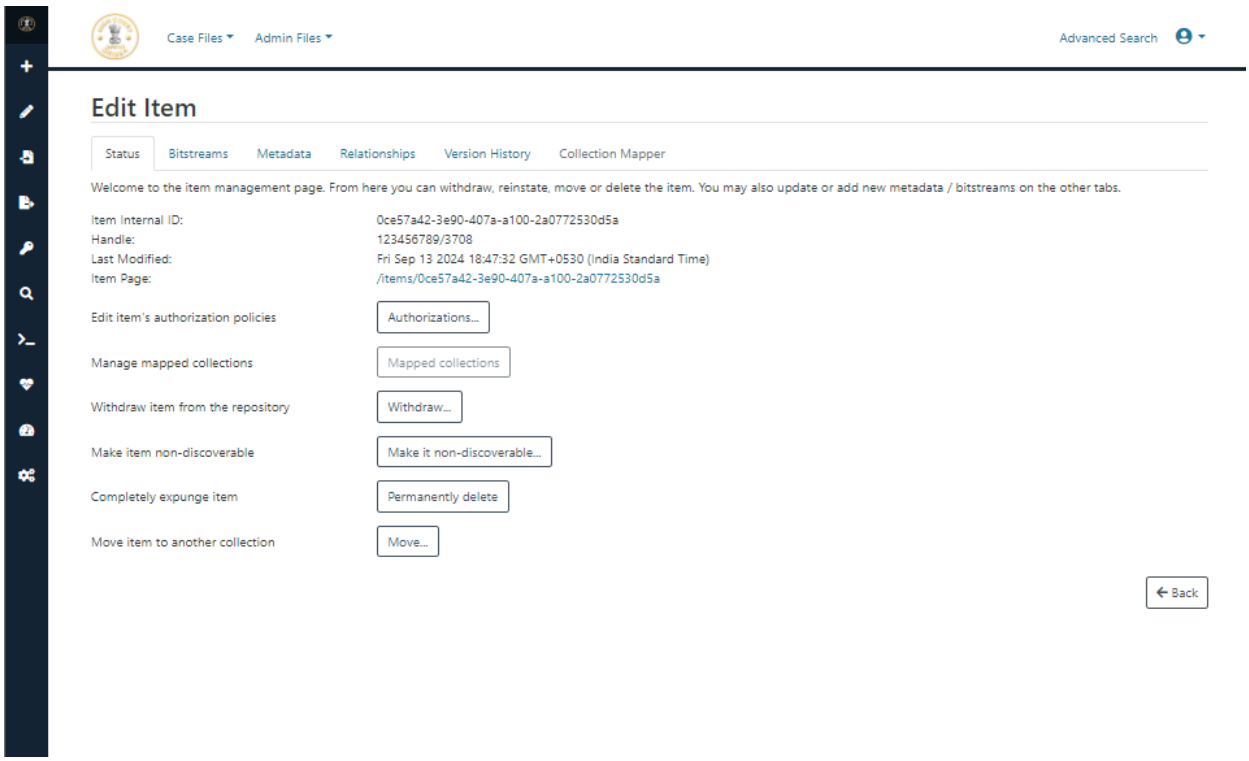


Here items can be edited and updated. For more information about editing please refer to Edit

metadata screen topic in subsequent chapters of this guide.

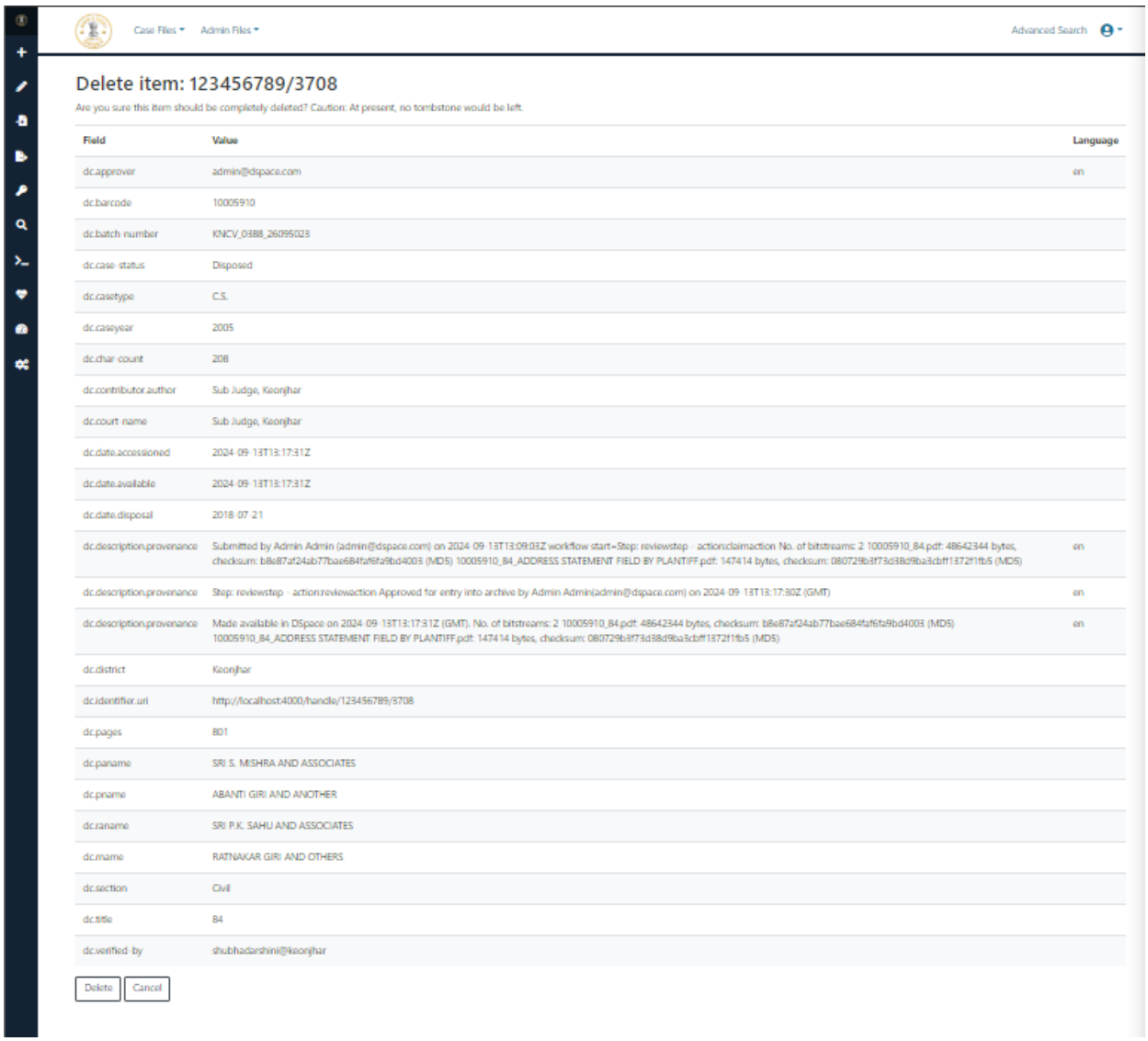
Administrator can withdraw, delete and move item to other collection. Following diagram shows

these buttons which are always above the item list.



When administrator clicks withdraw or delete button a conformation page is displayed to prevent

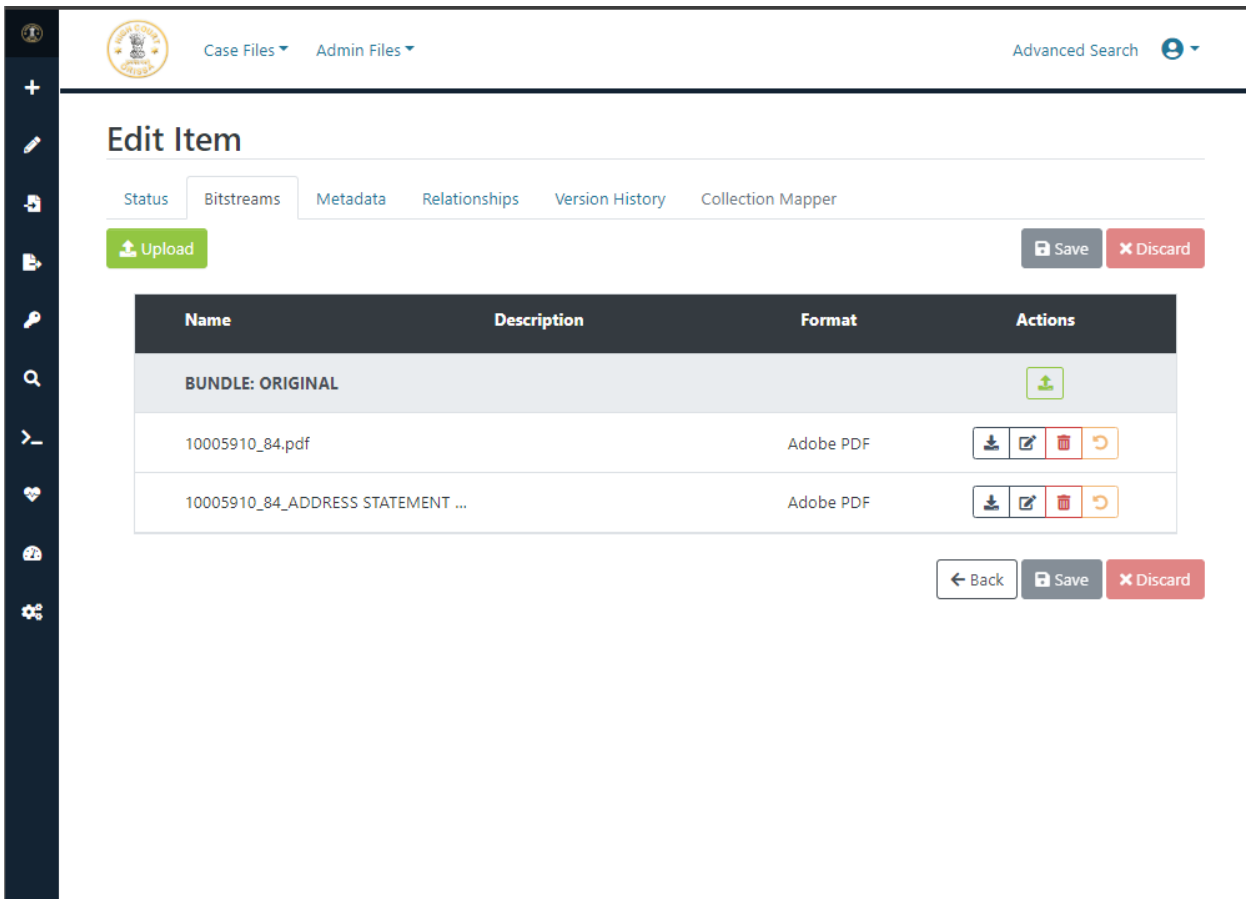
accidental data withdraw or delete.



Deleting bitstream

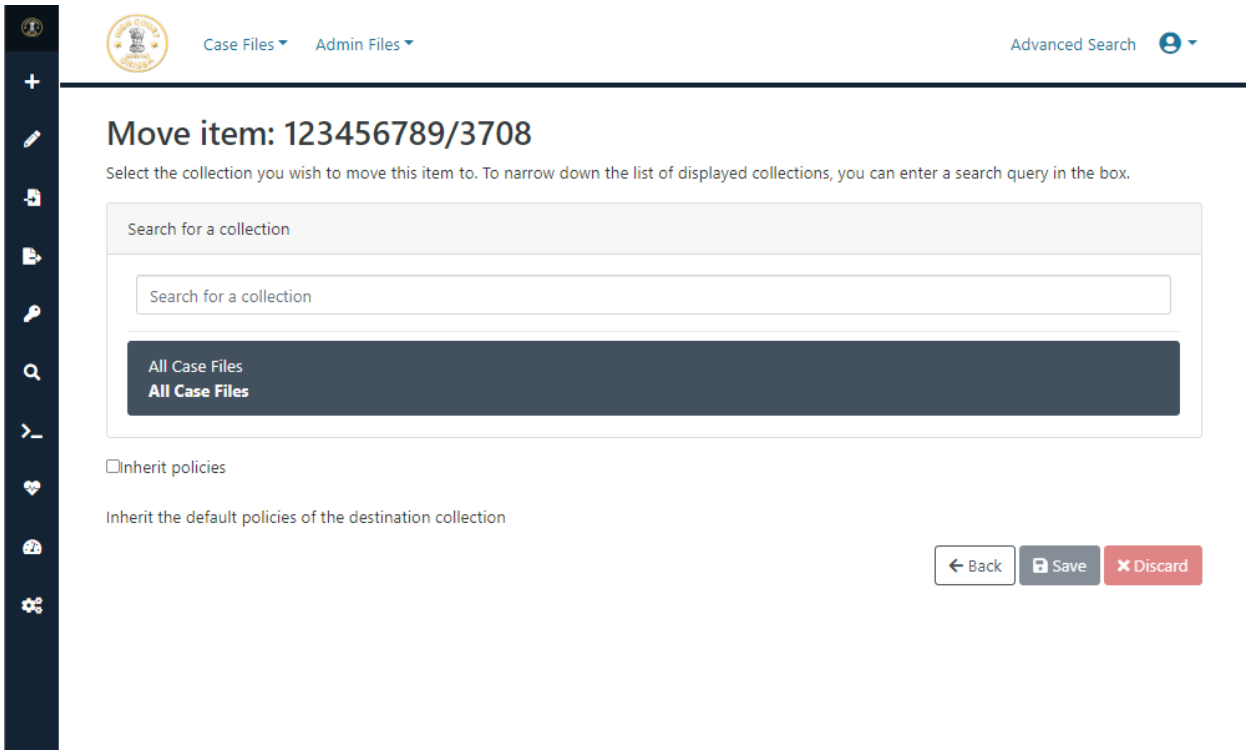
When you go to the Bitstream click on delete option to select the document and click on save

remaining Bitstream.



In move items a page indicating the new location of item is opened. On clicking move button the

item is permanently moved to the new location.

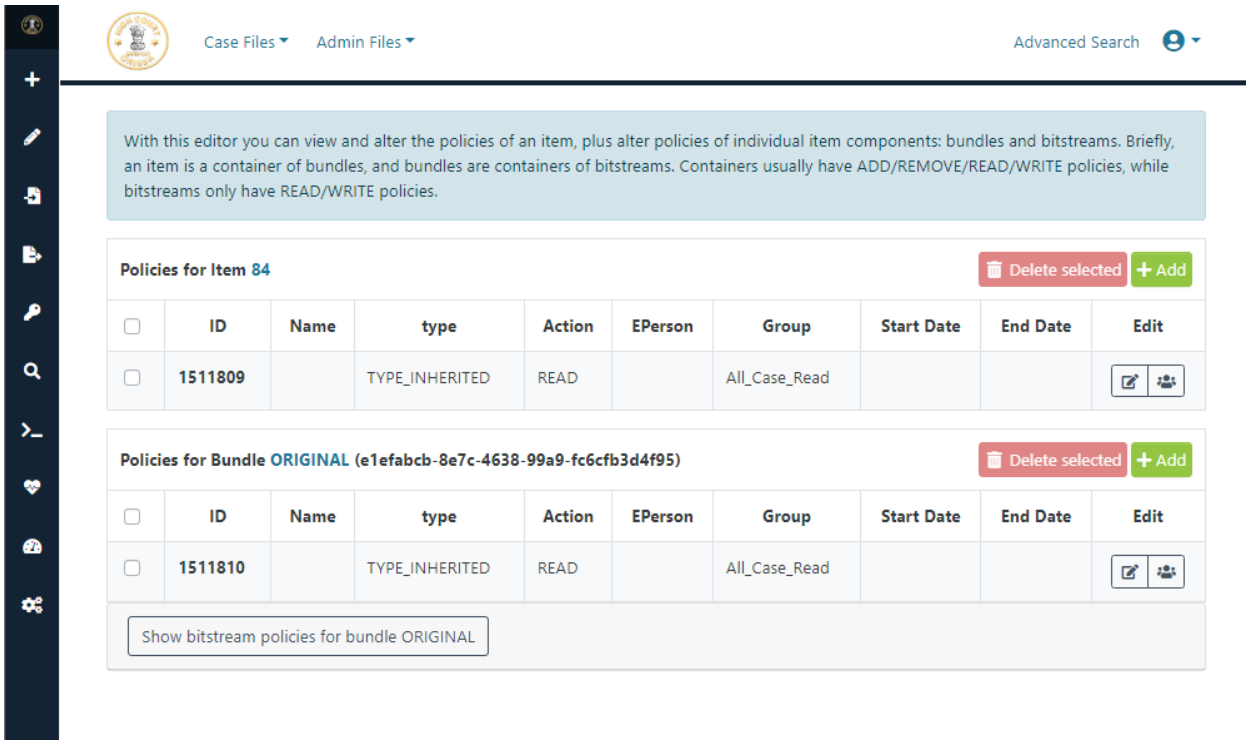


Edit button take cursor to the authorization tab. Following description gives a brief overview of

editing policies for an item in the authorization tab.

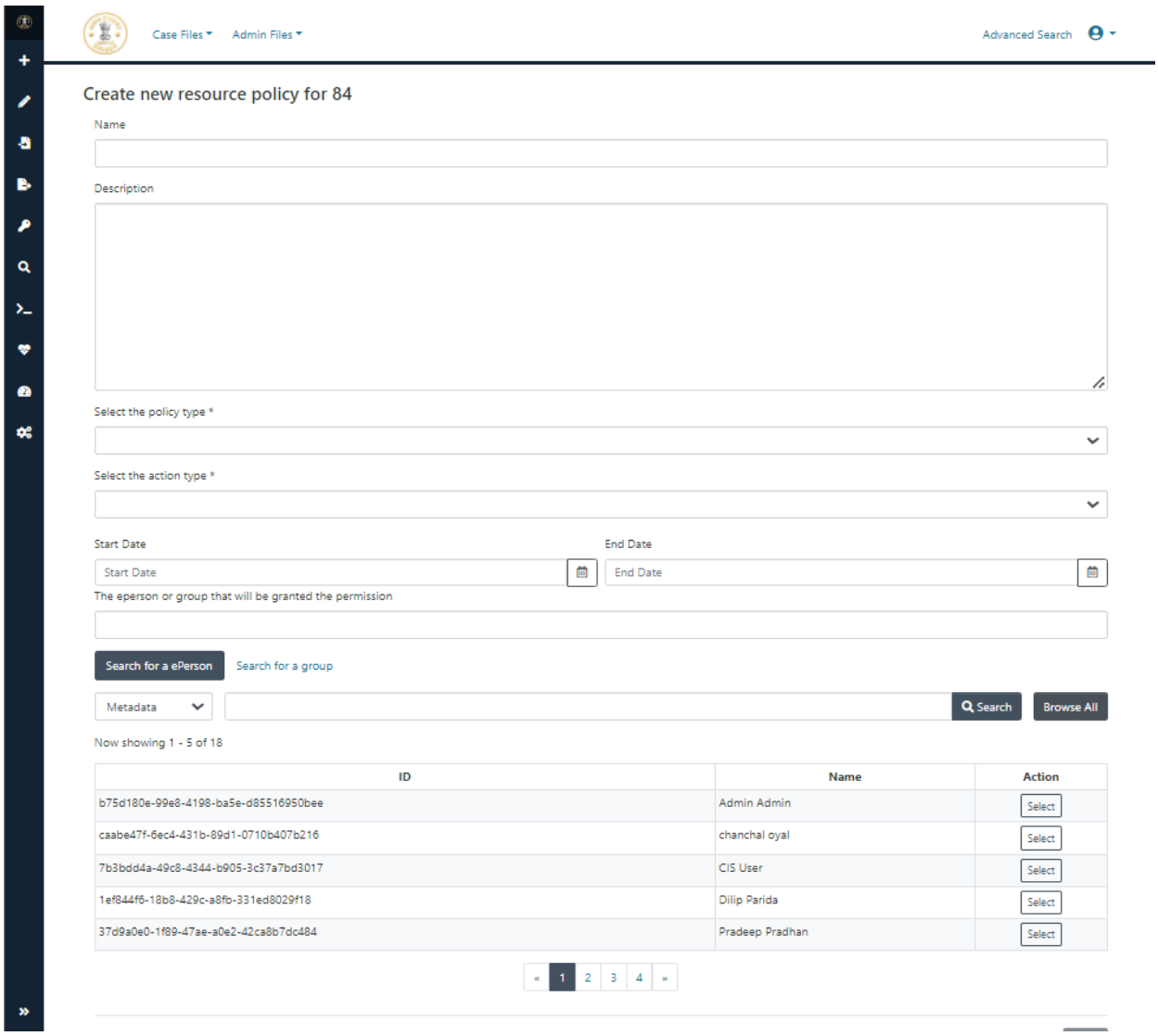
Administrator can alter the policies of item via edit button. Here administrator can edit, delete and

add new policies for bundles, bitstream and item as shown in the figure.



Under add new policy and edit button, an edit page opens where required details can be filled. With

delete button policy is deleted and changes are reflected in the present screen.



Step 8: User Activity & Access Tracking

In this Step, the focus is on implementing a robust auditing system that ensures transparency and security within the document management system (DMS). This helps in tracking user activities, preventing unauthorized access, and ensuring compliance with organizational and regulatory standards.

Key Features & Actions:

1. **Comprehensive Audit Log:**
    - Maintain a detailed record of all user interactions within the system.
    - Capture login attempts, user roles, and session durations.
    - Store timestamps, IP addresses, and device information for enhanced security.
2. **File Access & Modification Tracking:**
    - Monitor and log file access, including viewing, editing, and sharing actions.
    - Track modifications made to documents, including version history and rollback capabilities.
    - Implement notifications or alerts for unauthorized or unusual file access patterns.
3. **Download & Sharing Logs:**

- Record all instances of file downloads, including user identity, time, and file details.
- Track document sharing activities (e.g., external links, email attachments).
- Implement restrictions on unauthorized downloads, if necessary.

4. **User Role & Permission Auditing:**
    - Keep track of changes to user roles and permissions within the system.
    - Ensure accountability by logging administrative actions, such as granting or revoking access.
5. **Real-time Monitoring & Alerts:**
    - Set up real-time alerts for suspicious activities (e.g., bulk file downloads, repeated failed logins).
    - Provide dashboard analytics for system administrators to review audit logs efficiently.

Step 9: API Integration

This Step is centered around enabling seamless integration between the DMS and external systems, allowing clients and third-party applications to interact with the platform efficiently.

Key Features & Actions:

1. **Custom API Development for File Uploads:**
    - Develop secure and scalable APIs that allow clients to directly upload files into the DMS.
    - Implement authentication mechanisms such as API keys, OAuth, or JWT tokens.
    - Support bulk uploads, metadata tagging, and format validation.
2. **Push & Pull Functionality for Case Records:**
    - Enable API endpoints to retrieve (pull) case records from the DMS, allowing external systems to access necessary documents dynamically.
    - Implement push functionality, allowing case records to be sent to external applications in real-time based on predefined triggers.
    - Ensure APIs support various formats (e.g., JSON, XML, CSV) for data exchange compatibility.
3. **Security & Access Control:**
    - Enforce role-based access control (RBAC) for API users to prevent unauthorized data access.
    - Implement data encryption for secure transmission and storage of case records.
    - Introduce rate limiting and logging to monitor API usage and prevent abuse.
4. **API Documentation & Testing:**
    - Provide comprehensive API documentation for clients, including endpoint details, request/response formats, and authentication methods.
    - Develop sandbox environments for testing and validating API integrations.
    - Ensure APIs adhere to industry standards (e.g., RESTful architecture, OpenAPI specifications).

5. **Seamless Data Exchange & Automation:**
    - Enable automated workflows for case record synchronization, reducing manual intervention.
    - Support webhook-based notifications for real-time updates on file changes or case status.

Next Steps

After completing the above Steps, we will proceed with the following:

- ## 1. Supreme Court Storage Structure
- **Objective:** Implement a secure and scalable file storage system as per Supreme Court requirements.
- **Key Features:**
    - Hierarchical storage based on **case type, court level, and filing date**.
    - **Document retention and archiving** based on legal mandates.
    - **Role-Based Access Control (RBAC)** for judges, clerks, and legal professionals.
    - **Redundant backups & disaster recovery** to prevent data loss.
    - **Advanced search & indexing** with metadata tagging and OCR.

- ## 2. Wacom Integration (Digital Signatures & Annotations)
- **Objective:** Enable secure **digital signatures and document annotations**.
- **Key Features:**
    - **E-Signature for judicial documents** (orders, rulings, warrants).
    - **Handwritten annotations** for judges to add notes and highlights.
    - **ISO-compliant digital authentication** for legal validity.
    - **Biometric verification** with pressure-sensitive signature capture.
    - **Audit trail for signed documents** stored securely in the DMS.

- ## 3. Annotation & PDF Masking
- **Objective:** Enable **document annotations and secure redaction** of sensitive information.
- **Key Features:**
    - **Judicial Annotations:** Judges and clerks can **add notes, highlights, and comments**.
    - **Manual Redaction:** Select and mask **confidential case details**.
    - **Version Control:** Maintain both **original and redacted copies** of documents.
    - **Access Control:** Judges/lawyers access **unmasked documents**, while public versions remain **redacted**.

- **4. Database Modifications as per Supreme Court SOP**
- **Objective:** Align the database structure with **Supreme Court's Standard Operating Procedures (SOP)**.
- **Key Features:**
    - **Expanded case metadata** (case type, parties, hearing dates, document status).
    - **Version control & audit logging** for document changes.
    - **Scalable architecture** to store large digital evidence files.
    - **Seamless integration** with **Supreme Court DMS System**.

- **ISO Standard Compliance**: Compile and implement ISO standards as per Supreme Court SOP.
- **Fully Paperless Court Module**: Develop and integrate a paperless court workflow for end-to-end case management.
- **Workflow Automation**: Implement a structured workflow automation system to enhance judicial efficiency.